

**Design and Evaluation of an Agentic AI System for
Supporting International Students in Program
Applications to Germany Universities**

Master thesis

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II. Abstract

The application process for German universities is characterized by high bureaucratic friction, complex regulatory frameworks, and fragmented information sources. For international students, navigating the intersection of academic requirements and legal constraints often results in critical application errors. This thesis proposes an Agentic AI System designed to automate and optimize this decision-making process.

This system employs a modular architecture of autonomous agents, the system was evaluated using a "Human-in-the-Loop" methodology, testing five distinct applicant profiles against a production database of 106 Master's programs.

The results demonstrate that the system functions as a comprehensive strategic advisor rather than a simple retrieval tool. Key findings indicate that the system achieved accuracy in enforcing critical safety constraints, successfully distinguishing between high-risk profiles (e.g., Vietnam, India) and standard applicants (e.g., Europe Union) to generate legally accurate timelines. Furthermore, the system successfully distilled complex data into actionable "Application Strategy Dossiers" (as a PDF file) and precise Top-N Program Recommendations, balancing hard constraints with soft preferences.

While the evaluation identified limitations in deep-web retrieval regarding nested formatting requirements, the research confirms that Agentic AI can effectively democratize access to higher education. By automating the high-friction tasks of constraint checking and timeline calculation, the system provides a robust safety net for international mobility.

Keywords: Agentic AI, Large Language Models, International Education, Decision Support Systems, Automated Reasoning, German University Admissions.

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VI. List of abbreviations

EU European Union

DAAD Deutscher Akademischer Austauschdienst (German Academic Exchange Service)

AI Artificial Intelligence

LLM Large Language Model

LLMs Large Language Models

ECTs European Credit Transfer and Accumulation System

JSON JavaScript Object Notation

1 Introduction

1.1 Problem definition and Motivation

The percentage of international students pursuing higher education abroad has increased with 1.76 million students were undertaking across European Union (EU) while Germany, accounting for 24% in 2023 (Eurostat, 2024). Germany attracts students from all over the world due to the advantages of its education system, such as tuition-free public universities and strong academic reputation. By the winter semester of 2024/25, a total of 492.087 international students were enrolled at German higher education institutions. The share of foreign students has increased from 9.2 percent in 1998/99 to 17.2 percent in 2024/25 (Centrum für Hochschulentwicklung [CHE], 2025).

However, the application process for international students is often complex and time consuming. Requirements differ not only between student countries but also across universities. For example, students from non - EU countries often face additional requirements. Currently, students must gather information manually from multiple sources such as official university websites, aggregator platforms. They then need to filter options, compare requirements, take notes, create checklists, and set deadlines individually and check with the official sources to find the most suitable programs. This manual process could be prone to errors and easily overwhelm applicants.

Meanwhile, recent advances in Artificial Intelligence (AI) have transformed how people can leverage AI across industries. Particularly, the rapid evolution of Agentic AI, a new and trending paradigm in which AI systems operate proactively, orchestrate tools, and maintain memory to support complex workflows. Conventional AI models are typically designed for specific tasks and lack the autonomy to adapt dynamically to complex environments. Agentic AI as a category of AI systems capable of independently making decisions, interacting with their environment, and optimizing processes without direct human intervention

Based on that, the idea of design and evaluate an Agentic AI system could support students throughout the application process pop ups, it would not only simplify applying to master's programs in Germany but could also be extended to universities across EU and beyond or it could also be narrowed to specific University.

1.2 Research questions

Problems:

When preparing to apply for a University abroad, students must manually collect, compare, and check requirements across universities through many websites, maintaining their own notes and reminders for deadlines and documents. The normal application period often lasts from 2 to 3

months. But due to the additional requirements, the preparation time for international students could be much longer. Students from non-EU countries have earlier deadlines and shorter time to prepare. For example, students from Vietnam, China, etc. ... have to experience a knowledge test so called "APS" with "Germany Ambassador Office" and which only happens 2 times in a year. And most of Universities require non-EU countries to apply via a third party website so called "Uni Assist" for checking the valid documents which could take time to evaluate. These lead to inefficiency, high cognitive load, and an increased risk of missing important steps or missing information. There is currently no adaptive, and autonomous solution that dynamically tailors the application process to an individual student profile while also handling planning, monitoring, and proactive notifications to support students for their well preparation.

Existing AI tools such as AI-based assistants or aggregator websites provide only partial support for international students. These solutions are often static and fragmented:

- Aggregators present program information which is from many Universities but do not integrate profile-specific requirements or personalized recommendations.
- General-purpose AI tools can answer isolated queries, but they lack process awareness and cannot coordinate tasks across the entire application journey.

Research Question:

How can an Agentic AI system be designed and evaluated to support international students in applying to Master's programs in Germany?

Additional Research Questions:

RQ1: What requirements and challenges do international students face in the German Master's application process, and how can these inform the design of an Agentic AI system?

RQ2: How does the integration of rule-based analytics and LLM-based (Large Language Model) reasoning improve program matching, requirement summarization, and timeline management?

RQ3: What is the user-perceived value of the Agentic AI system in terms of efficiency, clarity, and time savings compared to existing tools?

1.3 Objectives

Design and implement a Proof of Concept Agentic AI system that supports international students throughout the Master's application process in Germany.

Evaluate the system's efficiency and effectiveness in comparison to manual processes.

Demonstrate the broader potential of Agentic AI in streamlining complex processes within education and other industries.

1.4 Structure of work

This thesis is structured into six chapters.

The first chapter introduces motivation, problems, research questions, and objectives, and positions the work within the broader context of Agentic AI for practical applications.

Chapter 2 presents the theoretical foundations and related research about Artificial Intelligence and Agentic AI. And it further discusses AI models (Large Language Models), and overview of Agentic AI workflow including the Thought-Action-Observation cycle, tools use, and Agentic Retrieval Augmented Generation. The chapter surveys relevant Frameworks, contrasts single-agent and multi-agent settings, and examines application domains and evaluation practices.

Chapter 3 outlines the research design following Design Science Research approach with a supplementary by Research Exploration by Expert Interviews. It specifies the data sources which is from DAAD as backbone and official University websites in conditions. It presents the system design and workflow, the implementation plan, and the methodology for evaluation. Ethical and legal considerations are articulated to ensure compliance and responsible research practice.

Chapter 4 documents the implementation of the system. It explains the choosing model reasons and describes the architecture, data handling and storage components such as the program catalog and profile catalog, the integration strategy with selected framework, the implemented modules, and the functional prototype for inputs and outputs

Chapter 5 reports the evaluations. It defines the evaluation setup, presents quantitative and qualitative results, and discusses the findings with respect to the research questions and objectives.

Chapter 6 provides the overall discussion and conclusion. It summarizes the main findings, reflects on limitations, proposes directions for future work, and states the final conclusions of the study.

Apart from 6 Chapters, at the end of this research paper including Appendix for list of Interview Expert questions, Glossary, Bibliography and Declaration of Authenticity.

2 Literature Review

2.1 Artificial Intelligence and Agentic AI

From history, AI focused on rule-based and human intervention, response to specific inputs and stimuli which is an early form of Classic AI. Classic AI were primarily rule-driven and depend on human instruction with predefined tasks in controlling environment. Their intelligence lay in executing explicit instructions rather than learning or reasoning.

Later than with the development of Large Language Models (LLMs), Generative AI (GenAI) represent a significant evolution which learn patterns from large scale number of data and can handle variety of data content such as number, text, audio, video. It essentially performs a reverse operation to Classic AI, which takes learned representations and combines them to make outputs that look like human creativity. Generative AI systems show great zero-shot generalization and are now common in both research and industry because they can be used in many ways and are easy to integrate. Although they result high quality or artifact but still lack of ability to engage in sustained planning, adaptation or self-correction over time (Acharya et al., 2025).

A next paradigm shift in Artificial Intelligence - Agentic AI emerge which is defined as an enabling systems to act independently, pursue broad objectives rather than isolated decisions, and carry out complex tasks that require reasoning elements such as planning and reflection (see Table 1).

Acharya et al. (2025) stated that Agentic AI goes beyond generative and rule-based paradigms with goal-oriented, input-informed, and adaptable characteristics that allow it to accomplish intricate and multilayered task. Within the AI field, Agentic AI serves as a different form of intelligence that can take more autonomously functioning agentic behaviors that AI is no longer limited to executing pre-defined tasks or following content-generating algorithms.

Table 1: Comparision Generative AI and Agentic AI

Aspect	Generative AI	Agentic AI
Reasoning	Provides immediate responses but may lack sustained reasoning or verification.	Performs iterative planning, reasoning, action and reflection through the Thought-Action-Observation cycle, enabling goal-driven decision making and self-correction
Interaction	Interacts with user in a single conversational turn.	Interacts with users, external tools, APIs, knowledge bases, and many AI agents within an environment, supports multi-turn, multi-modal exchanges

Execution Capability	Performs single-step or static tasks without persistence; each prompt initiates a new process.	Executes multi-step workflows comprising sequences of dependent actions that require coordination, monitoring, and adaptive branching
Adaptability	Can not improve or update behavior after deployment, bound to training data.	Learns from experience through memory, feedback, and contextual awareness, dynamically refines strategies and tools choices
Autonomy	User-driven and reactive, only respond to user when get prompt.	Autonomous and proactive, can initiate actions, replan goals, and collaborate with other agents

Table 1 is adapted from “*Table 1: Comparison of Generative AI vs Agentic AI characteristics*” by Schneider, J. (2025) which is described Generative AI and Agentic AI based on different aspects including Reasoning, Interaction, Execution Capability, Adaptability and Autonomy.

Follow this, an Agentic system which can support international student in application process would not merely answer student’s question such as “Which Universities, Programs suit me best” but actively memorize, plan, adapt, coordinate subtasks over time autonomously.

2.2 Agentic AI Components

2.2.1 Large Language Models

An AI Agent is divided into two main parts, we can call it basically: the brain and body. The brain (AI model) as its core usually LLM as the most common type of AI model which is for reasoning, planning and reflection. LLMs are trained from a huge amount of data which allow to pattern, structure and nuance in language, however LLMs are only understand and generate text from human language. Agent need Tools in order to perform any tasks to complete Actions. The body of an AI Agent refers to its built-in capabilities and External Tools used to complete actions. (Yu et al., 2025)

LLMs are typically built on transformer-based neural language models, they are trained on massive corpora via a next-token prediction objective, given by a sequence of previous tokens input. With the emergent abilities including in context learning, follow instructions and multi steps reasoning, after pretraining, they can be adapted using specialized techniques to behave as conversational, reasoning, or interactive modules to solve complex tasks (Minaee et al., 2024).

LLMs represent an advancement in Artificial Intelligence by expanding machine learning systems from static prediction to dynamic reasoning and content generation. Minaee et al. (2024) stated in

the survey a comprehensive overview of model families: training regimes including supervised, self-supervised, reinforcement fine-tuning; adaptation methods including instruction tuning, few-shot learning, reinforcement learning from human feedback; and the key benchmarks used to evaluate reasoning and linguistic coherence.

Basing on this overview, two kinds of open and proprietary models further diversify the LLM ecosystem recently including closed-source and open-weight models. GPT-4 (OpenAI, 2023) is such a common closed-source models which is accessed through APIs and can demonstrate exceptional reasoning and multimodal. In contrast, open-weight models such as Llama 3 (Meta AI, 2024), DeepSeek-R1 (DeepSeek AI, 2025), and SmolLM2 (Hugging Face, 2025) enable self-hosting, fine-tuning, and deeper experimentation through agentic frameworks such as LangGraph, AutoGen, or Semantic Kernel.

Table 2 summarizes several common LLMs and their providers, illustrating the variety of architectures and access types available for agentic AI research, the decision for implement Agentic system for this research topic will be further discussed in Chapter 4.

Table 2: Common Large Language Models

Model	Provider	Release Year	Access Type	Key Notes
Deep-Seek-R1	DeepSeek AI	2025	Open-source	Emphasizes efficiency and multilingual reasoning; suitable for lightweight deployment.
GPT-4	OpenAI	2023	Closed (API)	State-of-the-art reasoning and multi-modal generation; commercial benchmark for agentic workflows.
Llama 3	Meta AI (Facebook AI Research)	2024	Open-weight	Supports local fine-tuning; widely used as an open baseline for research and education.
SmolLM2	Hugging Face	2025	Open-source	Compact model optimized for on-device and embedded agent scenarios.
Gemma	Google Deep-Mind	2024	Open-source	Lightweight and efficient model optimized for reasoning and code tasks.

2.2.2 Tools

As mentioned in LLMs part, a key aspect of Agentic AI is taking Actions which is implemented through the use of Tools. Tools act as a function given to a LLM. LLM only has pre-train data act as an internal knowledge, therefore if an agent need to do more specific task, we must provide it through Tools. In many agentic workflows, the agent decides whether to call a Tool, composes parameters for it, then parses the results into its ongoing reasoning. Without such tools, an agent remains confined to its training timestamp and cannot fetch real-time data, execute actions, or interact with external systems (Li et al., 2024).

For instance, when student asked: “What is the current deadline for the Computer Science Master’s program at University X?”, the LLM must recognize the need to invoke a search or web-scraping tool rather than rely on stale internal facts. After invoking that tool, the agent will receive results, integrate them into its internal state, and continue planning accordingly. Wang et al., (2024) describe this pattern as foundational to the “action” aspect of agentic systems.

Furthermore, Platt et al. (2025) defined agentic LLMs by their ability to reason, act, and interact-where “act” necessarily implies tool invocation or interaction with external systems. Likewise, Wang et al. (2025) situated tool integration as a core technique in agentic systems, elaborating on APIs, execution pipelines, and monitoring.

2.2.3 Agentic Workflow

Agentic AI systems operate in a recurring loop of Thought - Action - Observation until the objective of the query is fulfilled.

- Thought: internal reasoning and planning.
- Action: executing operations or tool calls.
- Observation: receiving and processing feedback.

This Thought - Action - Observation cycle is iterative until the agent completes its target or transitions to a terminal state. The loop allows the agent to adaptively refine its strategy before giving a final answer rather than making a single, static prediction (see Figure 1).

There are 2 prompting techniques using in Thought which are Chain of Thought and React Approach. While Chain of Thought guide the model to think step by step, React Approach as Reasoning (“Thought”) and task-specific (“Action”) are in interleaved manner that incorporates environmental feedback (“Observation”). In this originate framework, the agent formulates an internal reasoning trace, executes an action such as a tool call or communication with external sources such as knowledge based or environment, then observes the returned result, and updates its context continuously before planning the next step (Yao et al., 2023).

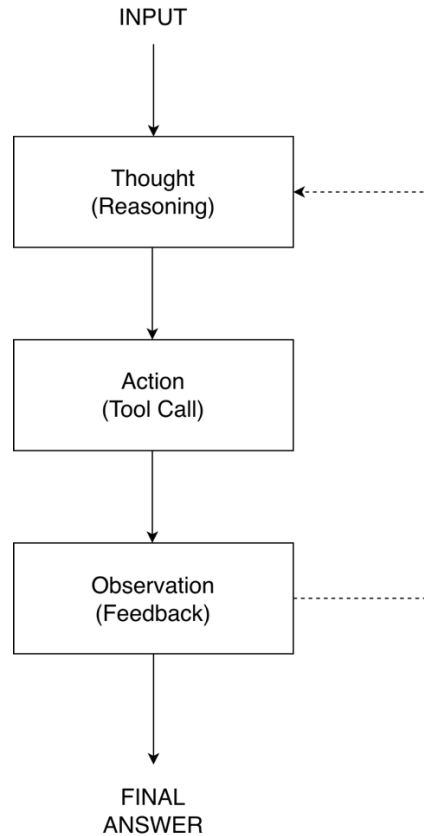


Figure 1: Thought - Action - Observation Cycle

Zhao et al. (2025) also described the Thought - Action - Observation process as the core workflow of an Agentic AI system, representing how Thought outputs are transformed into concrete Actions and evaluated through Observations. Similarly, Acharya et al. (2024) highlight that agentic AI development beyond static generation by continuously refining objectives and strategies through these cycles, achieving adaptive and self-correcting intelligence.

The detail approach of Thought - Action - Observation will be discussed further in chapter 4.

2.2.4 Frameworks

Agentic AI is deployed increasingly in many application industries, to support its capabilities, a growing ecosystem of agentic AI frameworks has emerged, some leading Frameworks such as CrewAI, LangGraph, AutoGen, and Semantic Kernel, Agno, Google ADK, and MetaGPT. The main purpose of those emerged Frameworks are orchestration tools and Agent workflow, they provide the architectural foundations and tooling necessary to build, orchestrate, and deploy intelligent agents that reason, act, and adapt autonomously.

Derouiche et al. (2025) stated that agentic frameworks function as the foundational layer for systems facilitated by LLM, wherein supporting agents demonstrate autonomy, contextual awareness, and goal-oriented behavior. They operationalize the agent's cognitive loop planning, take action by using

tools, receiving observations, and improving plans while ensuring persistence, safety, and coordination. Frameworks combine capabilities that allow agents to manage complicated tasks with minimal intervention from humans.

One of the important aspects of Frameworks also is systematic assessment, Framework is the essential parts of development and deployment process. They allows continous monitoring and in-depth error analysis, offer flexibility for developers to design and evaluate base on their own scenario. These frameworks are highly general and designed to support a wide range of development use cases instead of being limited to specific tasks. This makes them versatile tools for AI research and application (Yehudai et al., 2025).

Collectively, these studies position frameworks are essential to enable agentic behavior, integrating reasoning, memory, and tool use into cohesive systems capable of autonomous and adaptive performance.

Table 3: Comparison of Common Agentic AI Frameworks

Framework (Provider)	Core Focus	Key Features
LangGraph (LangChain/ LangSmith Eco- system (2024))	A graph-based novel model for sequencing tasks among LLM agents	Node-edge workflow enabling reasoning, memory persistence, and dynamic tool calls (nodes represent actions and Edges control flows between nodes).
CrewAI (Open-source Community (2024))	Role-based multi agents, emphasizing coordination and delegation	Opens source, multi-agent orchestration framework, take advantage of a specialized planning agent which will create a step-by-step plan for all tasks, and share it with the crew
Semantic Kernel (Microsoft (2023))	Integration of LLMs with external systems	Provides enterprise-grade orchestration with fine-grained control over planning, memory, and skill execution
LangChain (LangChain Inc. (2023))	Tool and memory orchestration	Foundational framework for chaining prompts and tools; integrates retrieval, APIs, and vector databases.
LlamaIndex (LlamaIndex (2023))	Retrieval-Augmented Generation	Empowers agents with capabilities for querying structured and unstructured data, offers structured data

	(RAG) and data indexing	connectors, query engines, and memory stores for augmenting LLM reasoning with external knowledge.
SmolAgents (Hugging Face (2025))	Lightweight agent execution and evaluation	A very simple library, modular framework optimized for efficiency; supports reflection, planning, and function-calling pipelines.

Table 3 adapted from “*TABLE IV: Compatibility of Agentic AI frameworks with core service computing functions*” by Derouiche et al., 2025 and official framework documentation present 6 leading frameworks which is published in recent surveys about agentic AI. Each provides distinct architectural and capabilities which illustrate the diverse strategies used to support agents with planning, memory, and tool-use capacities. The selected framework for this implementation will be explained further in Chapter 4.

2.2.5 Agentic Retrieval Augmented Generation (RAG)

LLMs are trained base on a large amount of data to learn general knowledge. However, their dependence on static training data constrains their capacity to address dynamic, real-time inquiries, leading to outdated or erroneous outputs. In order to enhance LLMs and intergrating real time data retrieval and up to date response, Agentic Retrieval Augmented Generation (RAG) emerge as a tool solution. It solves this problem by finding and retrieving relevant information and forwarding that to the LLM.

While traditional RAG systems include retrival for querying external data sources, augmentation for processing retrieval data and generation for combining retrieved information, Agentic RAG intergrates reasoning, planning, and interactivity into the retrieval generation cycle, RAG workflows, allowing agents to autonomously generate, refine, and execute retrieval strategies in order to expand RAG. This agentic behavior transforms the traditional RAG process into an iterative Reasoning - Action - Observation loop, where the agent evaluates retrieval results, adjusts queries, and re-synthesizes answers in context (Li et al., 2025).

Agentic RAG includes those core components: reasoning-enhanced query formulation, dynamic retrieval planning, iterative synthesis, memory-based feedback, and self-evaluation (Li et al., 2025). In the same way, Su et al. (2024) refer to this as "dynamic retrieval control," in which the model independently determines "When and How to query external data".

2.2.6 Single Agent and Multi-Agents

While single-agent operates autonomously decision-making for focused tasks but they still face limitations when handles with complex, multi-dimensional problems requiring distributed cognitive

load and diverse expertise. Multi-agent systems fundamentally address these constraints through enabling collaborative between a group of intelligent agents to solve complex tasks collectively, transitions from a single model to collaborative centric approach (Tran et al., 2025).

In another research definition, multi-agent facilitates the collaboration between two or more agents. The systems are managed by task decomposition which complex problems are partitioned into subtasks that agents can handle independently or cooperatively (Wang et al., 2025). This decomposition enables specialized agents to concentrate to solve specific problem components, reducing the learning space and facilitating more effective coordination while still maintaining overall workflow coherensively.

The dynamic nature of task decomposition represents a critical advancement which subtasks are continuously adjusted based on preceding outcomes rather than remaining static (Wang et al., 2025). This real time recalibration enables systems to accommodate unforeseen outcomes or newly acquired information during execution, thereby improving overall task-solving strategies through adaptive planning.

Although multi-agent system executes by distribution, it synthesizes each agent contributions into cohesive solutions through information-sharing protocols and collaborative reasoning mechanisms. This collective intelligence emerges from coordinated interactions, enabling multi-agent systems to tackle problems that exceed any individual agent's capabilities and demonstrating clear superiority for complex, real-world applications requiring extensive information synthesis and adaptive problem-solving (Tran et al., 2025).

2.2.7 Agentic AI Applications

Agentic AI nowadays can be applied in almost industries, demonstrating the practical and efficient of using the Artificial Intelligence, moving beyond simple automation to systems capable of autonomous decision-making, planning, and proactive execution of complex, multi-step goals with minimal human intervention. For example (Bandi et al. , 2025) summarize applications of Agentic AI across many domains including Healthcare, Transportation, Software, Military and Security, Manufacturing and Industrial, Smart Cities and Energy, Public Administration, Tourism, Finance, Retail, Education and Scientific Discovery Research (see Table 4)

Table 4: Summary of selected agentic AI applications

Domain	Applications	Key features	Relevant to this study
Healthcare	Diagnostics and treatment planning, autonomous	Healthcare decision support, diagnosis, and personalization.	Relevant for comparing reliability and trust

	health monitoring, Clinic Decison		requirements across domains
Education/ Learning	Personalized assistant systems, adaptive curriculum agents, research and management	Personalized and adaptive support, multi-agent systems, intelligent task execution	Demonstrates agent collaboration and adaptive reasoning, support for students, teachers
Industry/ Businesses	Autonomous supply-chain and retail agents, workflow optimization, autonomous operations, predictive analytics for operations	Self-organization, negotiation among agents, goal-driven autonomy	Provides context for industrial adoption of agentic frameworks

Table 4 adapted summary of selected agentic AI applications from “*Table 5 in The rise of agentic ai: A review of definitions, frameworks, architectures, applications, evaluation metrics, and challenges*” by Bandi et al. , 2025. Table 4 represents the applications of AI varies differents domains which is combined in 3 main domains: Healthcare which most used in Diagnostics and treatment planning, also support in decisions; in Education, Agentic AI is applied as a personalized assistant which is responsible for different tasks; in other Industry and Business, Agentic AI also demonstrate widely applications.

2.2.8 Evaluations

Agentic AI covers in most intrustries and domains, when assess the overall performance of an Agentic AI, the evaluation is based on Core Evaluation Metrics which is surveyed and summarized by Bandi et al. (2025). It is devided into 2 parts including quanlitative method which is alongside with number and qualitive method which is estimate by human interaction. The key metrics of the evaluation is decribed as:

- Correctness / accuracy: e.g. eligibility matching precision, document requirement compliance
- Workflow success rate: percentage of runs that complete all phases without error
- Tool invocation accuracy: how often tools are called correctly, and outputs are valid
- Latency and efficiency: execution time, number of LLM calls, computational cost
- Fault recovery / robustness: ability to recover from tool failure or API error
- Interpretability / traceability: how well you can trace agent decisions, inspect state, rollback
- Scalability / concurrency: performance under multiple parallel agents or many tasks
- Safety / constraint adherence: violation count of guardrails, user privacy, consistency

Because some phases may remain supervised (especially high-stakes steps), human evaluation is essential:

- Expert review: domain experts check output correctness (eligibility decisions, document drafts)
- User study: prospective students use the agent and rate usability, trust, helpfulness
- Error diagnosis tasks: present ambiguous or contradictory inputs and see how agent handles or asks clarifications

3 Research Method

3.1 Research Design

Research design is primarily constructive (Design Science Research) to get the output of an AI Agent that can support international student in choosing and application process. This is supplemented by Qualitative Methods (Expert Interviews) to validate the core logic, get more insight from Administration Office members, not only from the student point of view.

Design Science Research for implement and testing the AI Agent architecture.

Expert Interview are conducted to get depth insight about the difficulties and the evaluation process of Administration Office

3.2 Research Scope

To ensure clarity and feasibility, the implement will focus specifically on supporting international students applying for Master's programs at German universities. This narrowed scope provides a well-defined case study while keeping the system extensible to other countries and contexts in the future.

Core functionalities:

- Profile - Program matching
- Summarization of program requirements, checklists.
- Deadline tracking and personalized timelines.
- Proactive notifications (weekly digest, reminders).
- Assistant for general questions

3.3 Data Collection

3.3.1 Validation Data

Conducting a few interviews with University Administration Officer or International Office Members who involve in managing and evaluating Master's application focusing on common applicant mistakes, matching algorithms parameters and trustworthiness of AI tools.

Interview purpose:

To gain insights from Pforzheim university admissions about common applicant difficulties, information management practices, important criaterias for application and their perspective on AI-based tools that could support international Master's applicants.

Target Interviewee:

Admissions Officer / International Office / Faculty Program members involved in managing or communicating with international applicants.

Duration: 30 minutes each Interview

3.3.2 System Data

Building a Program Catalog as a backbone of database system. The DAAD "Study Program in Germany" search engine serves as program catalog for identifying available master's programs, general requirements, and essential university links. In addition, detailed program requirements will be extracted directly from the respective University Websites to ensure the highest level of accuracy for Matching algorithms.

The DAAD (German Academic Exchange Service) is an association of German institutions of higher education and their student bodies. DAAD is also the world's largest funding organisation for the international exchange of students and researchers (German Academic Exchange Service [DAAD], n.d.). Database from this organization is liable and general for all of international who would like to apply programs, scholarships and learn more information about education in Germany University.

Data collected from DAAD for the purpose used in this study scope only master programs in Germany

3.4 Workflow

Workflow:

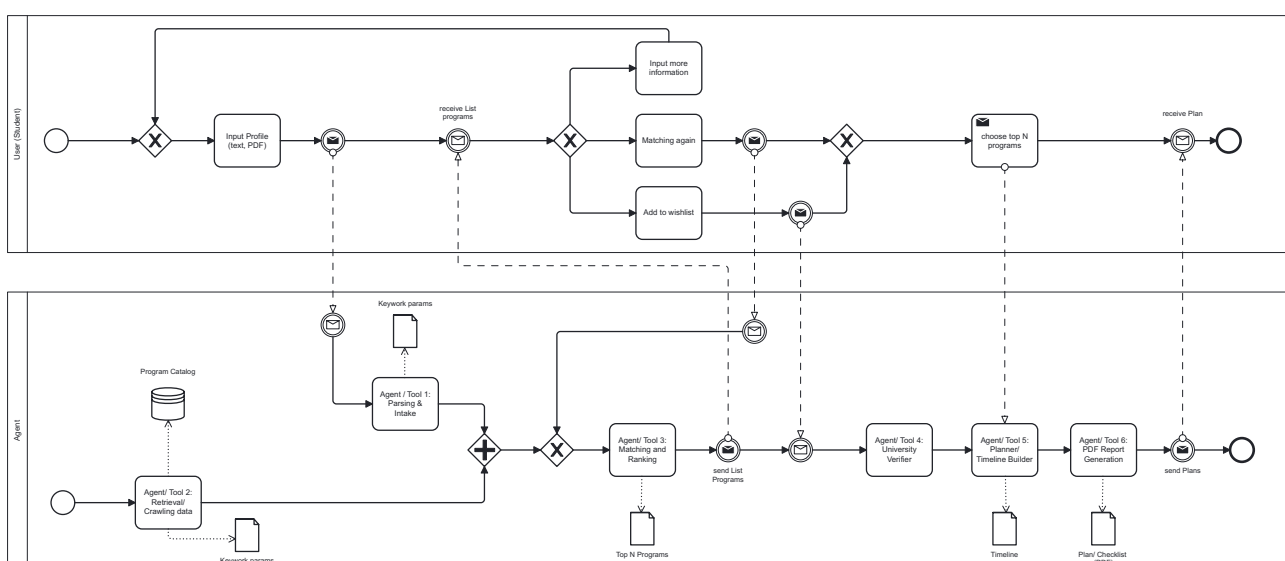


Figure 2: AI Agent Workflow (BPMN)

Agentic AI Orchestration:

The system will be implemented as an Agentic AI workflow, leveraging planning, tool-use, and memory capabilities. Candidate frameworks include LangGraph or LlamaIndex, in combination with LLMs and external APIs. The orchestration will enable the agent to interact proactively, adapt to user profiles, and provide personalized guidance across the full application process.

3.5 Evaluation and Analysis

The Agentic system will be evaluated by Agent-Level Task-Based. This framework treats the application process as a sequence of well-defined tasks and evaluates how effectively the agent can support their completion compared with a baseline AI assistant.

Tasks:

- Programs Matching: identify suitable master's programs given a profile.
- Requirement Summarization: extract all mandatory admission requirements.
- Timeline Building: generate a personalized application timeline.
- Assistant: answer general questions.

Metrics:

- Task completion rate: proportion of required outputs correctly identified.
- Time to completion: duration needed to finish each task.
- Critical error rate: missing or incorrect mandatory requirements.
- Re-plan frequency (telemetry): number of times the agent revises its plan or tool sequence.

Baselines:

- Agentic AI: the proposed system with planning, tool use, and memory.
- AI Assistant: a non-agentic baseline (e.g., ChatGPT in multi-turn Q&A).

4 Implementation

4.1 Requirement Analysis

The analysis requirement is based on the expert interviews conducted (see Section 3.2) in order to address real-world administrative challenges from University insights. Participants are six experts (5 interviews) who involving processing applications (Faculty Programs, Research Assistant, International Office, International Coordinator and StudiCenter). The key findings are five critical design imperatives: handling high-volume repetitive workflows, eligibility meets the general formal requirement, formal pitfalls, data reliability and ensuring data trustworthiness through verification.

4.1.1 Handling high-volume repetitive workflows

Regarding the volume of general inquiries which are clearly presented on the University website, all of the experts from different departments agree that its a huge workload and overwhelming . Expert A, B & C (Faculty Programs) and Expert E (International Coordinator) reported that 80% to 100% of student inquiries are repetitive, focusing primarily on "Am I eligible?" and document lists. Expert E noted that answers often consist solely of providing links to the website. Furthermore, Expert E and Expert F highlighted the challenge of time zones from different countries which could lead to delays waiting for simple answers.

Design Implications:

The process of answering general queries could be automation by AI system using RAG. It must autonomously resolve the "repetitive" questions (identified by Expert A, B & C and E) to free up human advisors for complex cases and provide general information about application process for student.

4.1.2 Semantic Matching vs. Keyword Search

Experts B and C (Faculty) identified a critical flaw in current manual search methods: the discrepancy between degree titles and actual curriculum content. They emphasized that a Bachelor's degree name in a non-EU country might differ significantly from the German equivalent, yet the field of study remains relevant. Expert E added that students are often confused between "ECTs requirements" and "Course names" and the "Bachelor's degree name".

Design Implications:

A simple keyword match is insufficient. The Matching Engine (Tool 3) must utilize LLM-based reasoning to analyze the content of module descriptions (semantic matching) rather than relying on exact title matches.

4.1.3 Mitigating Formal Pitfalls

There is a consensus between experts about the common mistake is missing document and common questions they usually get from student are about eligibility to apply for a specific program regarding the first bachelor degree in another countries and other documents. Specifically, applicants from non-EU countries often miss country-specific mandatory documents, such as the APS certificate for some specific countries. Furthermore, Expert D noted that applicants frequently perform incorrect GPA conversions, necessitating a manual "pre-check" by the university.

Design Implication:

A simple deadline tracker is insufficient. The Timeline & Checklist Builder (Tool 5) must be context-aware and combine with specific requirement. It is designed to generate dynamic checklists based on the user's origin country (triggering "APS" warnings) to prevent the specific incompleteness issues cited by the experts.

4.1.4 Data Reliability

To build a dependable Knowledge Base, the interviews examined which data sources universities consider authoritative. Expert D (International Office) established a clear hierarchy of trust, identifying Anabin (for university recognition) and HRK (German Rectors' Conference) as the internal sources of truth for verification reconized University in Germany. For program information, all of them confirmed that external platforms like DAAD and MyGermanUniversity are reliable partners. Expert A and Expert E explained that DAAD profiles are typically updated only once or twice a year via a manual review process and requirements do not change frequently.

Design Implication:

Although the requiremnts do not change frequently, but the system cannot treat aggregator data as real-time. The Retrieval (Tool 2) is designed to use DAAD as a foundational catalog but must cross-reference critical deadlines or requirements against the university's live website (via Tool 4: University Verifier) to bridge the update gap identified by Experts A and E.

4.1.5 User Acceptance and Trust Prerequisites

The experts expressed a cautious willingness to adopt an AI agent, contingent on specific trust factors.

Perceived Value:

The strongest driver for adoption is workload reduction. Expert A and Expert E explicitly hoped the system could handle the "80-90%" of repetitive traffic, allowing staff to focus on complex cases. Expert E and Expert F added that such a tool would solve time zone delays, providing instant

answers to international students who currently wait long periods for simple email replies. Expert B and C emphasized that true value lies in semantic matching which can identify eligibility based on course content and earning ECTS rather than just degree names.

Barriers to Trust:

Trust is not yet established. Expert F stated they do not yet trust AI due to accuracy concerns, identifying this as the single biggest barrier. Experts B, C, D, and E unified around Data Privacy (GDPR) as a non-negotiable requirement, specifically expressing concern about students uploading CVs or personal data to an external cloud.

Design Implication:

To secure expert recommendation (as promised by Experts E and F if the system is proven effective), the architecture must prioritize Transparency and Evaluation. The system is designed to provide citations for every answer (addressing Expert F's accuracy concern) and minimize data retention (addressing the GDPR concerns of Experts B, C, and D).

4.1.6 Summary key findings

The following table summarizes how the specific insights from the six experts translated into the system's core tools:

Table 5: Summary Interview's key findings

Key findings	Sources	Corresponding System Feature
80-100% of questions are repetitive (about Eligibility)	All experts	AI Assistant (RAG) for 24/7 FAQ resolution.
Degree titles differ; Need to double check with module/course description to identify minimum requirements	Expert B, C	Semantic Matching Engine (LLM-based).
DAAD as a reliable data source	Expert A, B, C, D, E	Data from DAAD as a Program Catalog
Specific failures: missing documents	All experts	AI Assistant (RAG) for 24/7 FAQ resolution, Checklist Builder
Need for high accuracy and evaluation	Expert D, F	Evaluation process

The expert interviews showed that most questions from international applicants are repetitive. These questions are already available on university websites, but students do not check them thoroughly.

Many applications are rejected because students miss documents or do not follow specific requirements. Some students also make mistakes in ECTs conversion and do not understand special rules for their country, such as APS. DAAD as a reliable sources and data is being updated by University once to twice a year and not change frequently which can be the base for treating DAAD as the main Program Catalog. Experts said that such an AI system is useful if it gives accurate information and follows data protection rules. They believe the system can reduce workload and save time for both students and university staff.

4.2 System Architecture

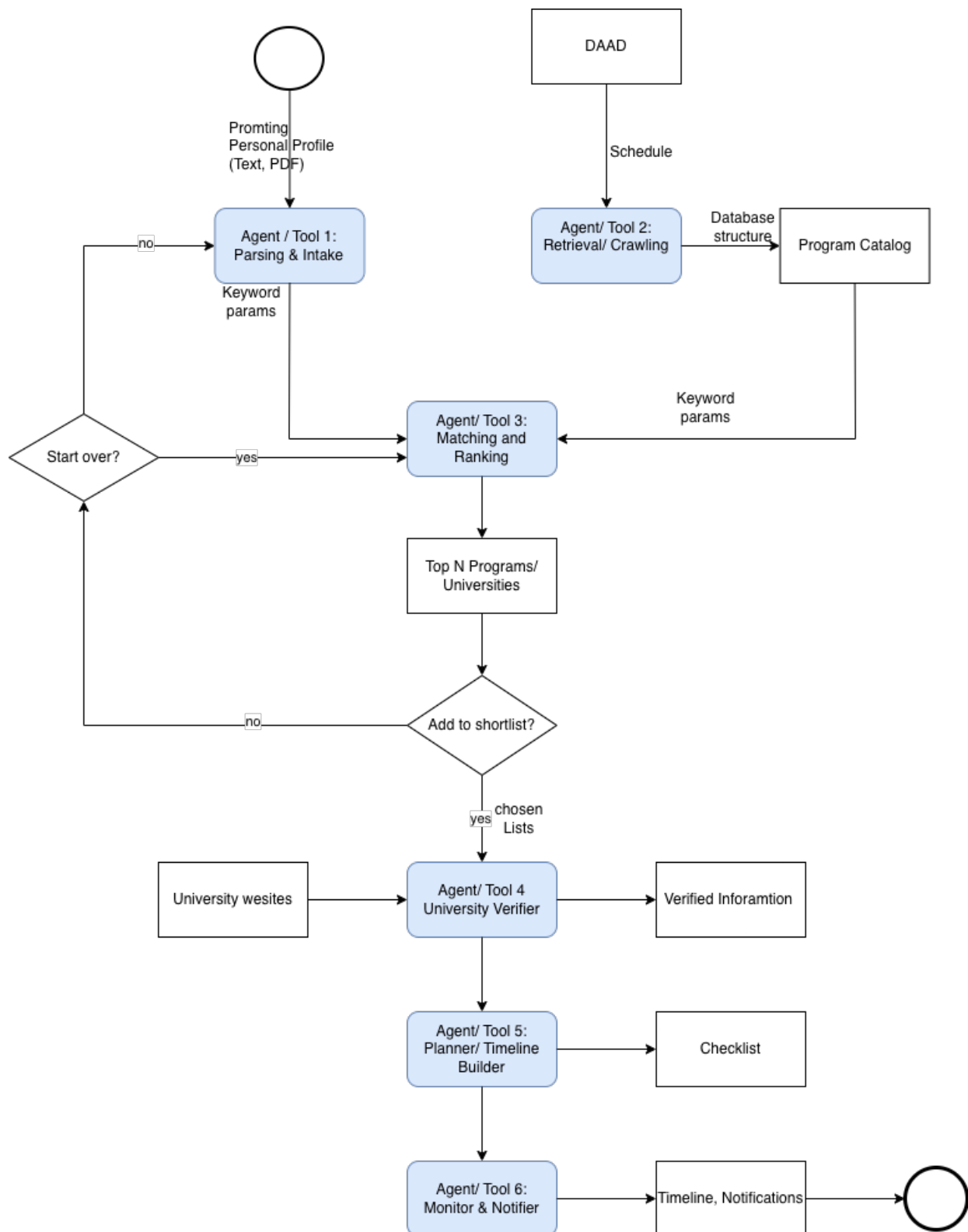


Figure 3: System Architecture

The proposed solution is designed as a modular that orchestrates multiple specialized agents to guide international students through the university application process. Unlike traditional chatbots

that rely solely on pre-trained knowledge, this system utilizes a Retrieval-Augmented Generation (RAG) framework combined with agentic tool use to perform distinct tasks: parsing, matching, verifying, and planning. The system workflow (see Figure 3) is orchestrated using a state machine that manages the flow of information between agents and human decision points.

4.2.1 Module Description

The system is composed of six distinct agents, as showed in the architecture diagram:

1. **Agent 1: Parsing & Intake**

Role: Constructs a structured UserProfile object.

Input: Natural language chat and uploaded PDF documents (Transcripts of Record).

Uses Gemini 2.5 to get key academic background fields and student's preferences. The GPA and ECTs then are normalized to convert to the German scale. It handles the interaction through conversation, asking follow-up questions if mandatory fields are missing.

2. **Agent 2: Retrieval & Crawling**

Role: Builds and maintains the static Program Catalog.

Input: Aggregated data sources (DAAD database).

This runs as a scheduled batch process, it structures raw data into a searchable JSON format, serving as the Long-Term Memory for the system.

3. **Agent 3: Matching & Ranking**

Role: Filters the all of programs in the Program Catalog down to a relevant Top N.

Input: User's Profile (from Agent1) and Program Catalog (from Agent2).

Implements a 4-Layer Filtering Funnel to get the ranking Top N Programs.

4. **Agent 4: University Verifier**

Role: Validating information for the user's shortlist.

Technology: Integrates with the Perplexity API to perform web searches.

It navigates to the official university website to extract the necessary information about the documents. For example, checks for country-specific mandates (e.g., APS Certificates for Vietnam/India/China).

5. **Agent 5: Planner & Timeline Builder**

Role: Synthesizes verified data into an actionable plan.

Input: Verified Information (from Agent 4) + User Profile (from Agent 1).

A deterministic logic engine that calculates backward from the deadline. It inserts buffer times for application processing and document notarization based on the user's citizenship.

6. Agent 6: PDF Report Generation

Role: Extract Application Strategy Dossier in PDF format.

Logic: Consolidate the intermediate outputs produced by Agents 1 through 5 into a persistent artifact: the Application Strategy Dossier in PDF format.

The detail of processing each module is described in the next section.

4.2.2 Core Technologies

- Large Language Model:

Google Gemini 2.5 The core reasoning engine for the agents is Google Gemini 2.5 Flash. This model was selected for reasons:

Long Context Window: Essential for Agent 1 to ingest and parse lengthy PDF transcripts without truncation.

JSON Mode Reliability: Critical for the "Intake" and "Matching" agents to output structured data (pydantic objects) rather than unstructured text, ensuring downstream code can process the results.

Efficiency: The "Flash" variant offers the low latency required for real-time interactive chat loops (Agent 1) while maintaining high reasoning capabilities for degree compatibility checks (Agent 3).

- Orchestration Framework:

LangGraph: The system logic is implemented using LangGraph, a library for building stateful, multi-actor applications with LLMs. Unlike simple linear chains (e.g., LangChain SequentialChain), LangGraph allows for:

State Persistence: A shared AgentState object passes data (User Profile, Program List, Verified Data) between nodes, ensuring that Agent 5 has access to the exact data verified by Agent 4.

4.3 Data Handling and Storage

The foundation of the Agentic AI system is the structured knowledge base. As described in the system architecture, the agent requires accurate, reliable and machine-readable data regarding university master programs to perform matching and ranking suitable programs for student. This section detailed the pipeline to retrieve, clean and transform unstructured DAAD content to structured "Program Catalog". This plays as a backbone database of the agentic system.

The data pipeline consists of two stages: (1) Data Extraction (2) Semantic Transformation using Large Language Models (LLMs).

4.3.1 Data Extraction

To form the initial dataset, a custom web crawler was implemented using Python and the Selenium automation framework. As identified in the methodology section and with the supplement about the reliable website, the data source selected was from the DAAD. Data is collected in the scope of this thesis as "International Programmes" which include all English master programs in Germany.

Data Crawling Logic (*Crawling_Data.py*) simulates a user navigating the DAAD portal. To ensure comprehensive data collection, the system implements a dynamic pagination handler. The crawler identifies total search results (programs) and iteratively navigates through pages by detecting and interacting with the "Next" pagination control button. To maintain stability and respect server load, randomized delays were implemented between requests.

Attribute Extraction for each program identified, the crawler extracts specific unstructured text fields required for the advisory agent. The following key attributes are retrieved:

- Program Metadata: Program Name, Institution, City, URL.
- Tuition fees: Tuition fees and semester contributions.
- Program Description
- Academic Requirements: Admission requirements
- Language requirements
- Application deadlines
- Submission channels.

In this stage, the system cross-references extracted cities against a predefined list of locations in Baden-Württemberg (e.g., Stuttgart, Heidelberg, Pforzheim). Only programs located within this state are persisted for further processing. This reduces noise and aligns the dataset with the initial validation scope defined in the thesis proposal. The belowed picture shows the initial JSON after crawling data.

program_id:	4722
name:	Actuarial and Financial Mathematics
institution:	RPTU University of Kaiserslautern-Landau
city:	Kaiserslautern
url:	https://www2.daad.de/deutschland/studienangebote/international-programmes/en/detail/4722/
tuition_fee:	None
semester_fee:	Social fee: 314 EUR per semester (including public transport ["Deutschland-Seme...
start_date:	Winter and summer semester
description:	Within the field of mathematics, the study programme focuses on financial mathem...
admission_req:	Applicants have to hold a qualified BSc degree in Mathematics or Economathemat...
language_req:	Profound knowledge of the English language (e.g., TOEFL 550 paper-based/ 213 co...
application_deadline:	Winter semester: 30 April (Early bird admission: 15 November) Summer se...
submit_to:	Graduate School "Mathematics as a Key Technology" Department of Mathematics RPTU U...

Figure 4: Initial JSON after crawling data

4.3.2 Semantic Transformation using Large Language Models

The initial data extracted through DAAD scraping is not well unstructured and inconsistent (Figure 4). For instance, the deadlines mixed with descriptive text, program description and language requirements described in different style and varying natural language formats. Especially for program description and program requirement which are two of the richest data to identify what program offer are different. Some of them may include a short description course name, some of them have a long description of course content, however in general, they all mentioned about the summary each program. Therefore, LLM-based transformation pipeline was implemented in order to make the data more structure and consistent as it convenient for matching in later stage. It can also extrac in a nice format and consistent regarding requirement summarization, program description which can be address Secondary Research Question 2 (RQ2).

In order to do that, a predefined JSON Program Catalog structure is also taken into account to make sure it will work well with student's profile.

The script *build_database.py* utilizes the LangChain framework to orchestrate a transformation workflow. The system employs Google's Gemini 2.5 Flash model to parse raw text into a strict JSON schema.

The transformation process involves prompting the LLM with specific extraction rules:

- Since the domain course is important to identify the best fit of bachelor program and master program, the specific ECTS requirement rule was set. The ECTS Normalization: The model

identifies credit points associated with specific subject areas (e.g., "16 CP in Mathematics") and structures them into nested objects.

- **Language Proficiency Standardization:** Natural language descriptions (e.g., "IELTS 6.5 or equivalent") are normalized into standardized objects containing the test name, minimum score, and Common European Framework of Reference for Languages (CEFR) levels.
- **Deadline Logic:** Since the deadline is different between EU and non-EU countries, it is divided into separate components. The model parses complex date descriptions (e.g., "July 15 for non-EU, Sept 15 for EU") and separates them into distinct *start_date* and *end_date* fields for different applicant groups and start semester.
- **Schema Validation:** To ensure the output is reliable for the Agentic AI, the output is parsed via Pydantic models. This enforces strict typing; for example, ensuring that *tuition_fees* are numerical and deadlines follow a specific nested structure.

The below picture shows a well-structured and consistent data set between programs after doing LLM

program_id:	7211
university_name:	University of Tübingen
program_name:	Advanced Quantum Physics
degree_type:	Master
application_mode:	Direct
city:	Tübingen
state:	Baden-Württemberg
tuition_fee_per_semester_eur:	1500
semester_contribution_eur:	0
min_gpa_german_scale:	2.5
required_degree_domains:	[1 items]
specific_ects_requirements:	[1 items]
required_standardized_tests:	[0 items]
min_work_experience_months:	0
requires_internship:	false
english_requirements:	{3 keys}
german_requirements:	{3 keys}
deadlines:	{2 keys}
course_content_summary:	Master's programme in Advanced Quantum Physics including experimental...
url:	https://www2.daad.de/deutschland/studienangebote/international-programmes/en/detail/7211/

Figure 5: Structured data after semantic transformation

The processed data is stored in a JSON document store (*structured_program_db.json*). This format was chosen for its flexibility and native compatibility with the Retrieval-Augmented Generation (RAG) tools used by the AI agent.

The final data model represents a significant enrichment over the raw source data.

4.4 Agent1: Parsing and Intake User's Profile

The first component of the system, Agent 1 is implemented to parse and intake user's information. This is a critical information to identify "Who is the student" and "What does he/she currently have". Instead of static form-based inputs, this agent utilizes a conversational state machine to iteratively build a comprehensive UserProfile. This agent is responsible for ingesting unstructured user data ranging from natural language chat inputs to PDF academic transcripts of record and transforming it into a structured, machine-readable format. The logic of converting grade and ECTs is also implement in this agent. The implementation leverages LangGraph to manage the flow between data extraction, logic application, and user dialogue.

4.4.1 State Management and Data Schema

To ensure consistency across the agentic workflow, the system employs a strict data JSON schema defined using Pydantic models (*models.py*). The agent's memory is managed via a global AgentState object, which persists across the conversation lifecycle. This state holds three critical components:

- Raw Intent: The user's unstructured queries or document uploads.
- Structured Profile: A validatable JSON object representing the student's academic and professional history.
- Interaction History: The conversational context required for the Large Language Model (LLM) to maintain continuity.

The data model captures all neccessary attributes for the German university application which could use to match to our Program Catalog. Key structures include:

- AcademicBackground: Through user input, it captures bachelor degree, country origin (essentail for identify EU and non-EU), total semesters, total credits earned, desired master program, field of interests, ..
- Domain knowledge: Through Transcript of Records, it captures the nested of all course name, responsive credit, ..
- Optional: User's preference to identify the maximum tuition fee that student can afford, prefer cities, states, teaching language, ..

However, as mentions in the problem of EU and non-EU contries, student come from differenent countries and university have different grading system and credit point. So, it is neccessary to do an additional step for converting those.

4.4.2 Rule-Based Normalization (GPA and ECTS)

The Modified Bavarian Formula (GPA):

To standardize international grades to the German 1.0 - 4.0 scale, the agent applies the following logic automatically upon extraction of grade metadata:

$$x = 1 + 3 \frac{N_{\max} - N_d}{N_{\max} - N_{\min}}$$

x : German Grade equivalent.

$N_{\max}$: The highest attainable grade in student's University grading system.

$N_{\min}$: The lowest passing grade in student's University grading system.

N_d : The grade that student achieved (to be translated into the German system).

ECTS Credit Conversion:

A major barrier for international students is the variation in credit systems. The agent implements a heuristic to estimate the European Credit Transfer and Accumulation System (ECTS) value for total credit points and every course in the transcript. The system calculates a dynamic conversion factor based on the standard European load of 30 ECTS per semester:

$$F = \frac{30 \cdot d_{\text{semesters}}}{C_{\text{total}}}$$

F : Conversion factor

$d_{\text{semesters}}$: Total semester that student studied in their degree program

C_{total} : Total credit points that student earned in their degree program

This factor is then applied to individual course and module by multiply with the original credit points so that the Matching Agent can query specific requirements (e.g., requires 10 ECTS in module Business Administration) regardless of the original grading system.

4.4.3 Agentic Control Flow

The execution of Agent 1 is governed by a State Graph (implemented via LangGraph) rather than a linear script. The control flow logic (*check_for_completion*) acts as a "Gatekeeper" with the following recursive states:

1. Parsing: The agent attempts to extract structured data from the latest input.
2. Validation: The *get_missing_fields* function scans the profile for mandatory attributes (e.g., Citizenship, Transcript, GPA).
3. Feedback Loop:

If mandatory fields are missing, the graph routes to a Chat Node, which prompts the user specifically for the missing data.

If the profile is complete, the graph transitions the state to the Matching Agent (Agent 3).

This architecture ensures that the system works proactively, guiding the user until a viable profile is constructed before attempting to search the database.

4.5 Agent3: Matching and Ranking

Once a structured UserProfile is generated by Agent 1, the workflow transitions to Agent 3, the "Matching and Ranking" module. This component is responsible for querying the Program Catalog and matching with student's profile to identify suitable Master's programs.

Agent 3 implements a Multi-Stage Filtering Funnel (see Figure 6) which goes through four layers. This architecture reduces the search space using increasingly sophisticated (and computationally expensive) techniques, ensuring that the heavy processing power of the LLM is reserved only for the most viable candidates.

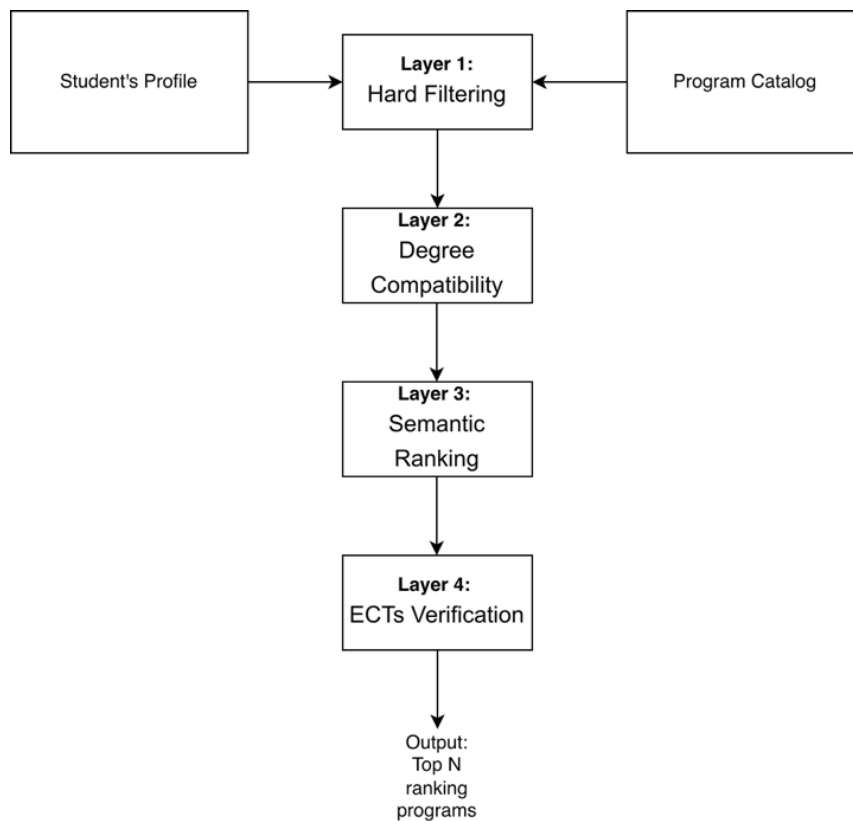


Figure 6: Multi-Stage Filtering Funnel (Agent3)

The detail implementation will be explained in the below sections.

4.5.1 Layer 1: Deterministic Filtering (Hard Constraints)

The first layer acts as a coarse filter to instantly eliminate programs that are technically incompatible with the user's profile. This layer utilizes strict boolean logic rather than AI probability. The checking constraints function evaluates specific criterias:

- GPA Threshold: Compares the applicant's calculated German GPA against the minimum required GPA of the programs
- Total Credits Point: Ensures the applicant's Bachelor's degree meets the minimum total ECTS requirement
- Student's Preferences:
 - Financial Situation: Filters out programs where the tuition exceeds the user's preference.
 - Geography: Enforces city and state preferences defined in the user profile.
 - Work Experience: Rejects programs (e.g., MBAs) requiring professional experience exceeding the user's working experiences.
 - Semester Intake: Removes programs that do not offer an intake in the user's preferred semester (Winter vs. Summer).

This step significantly reduces the dataset size at negligible computational cost.

4.5.2 Layer 2: Degree Compatibility

The Degree compatibility between bachelor's degree and master's degree is an important factor for matching. That is the reason why almost master program mentioned in their first requirement about "required bachelor degree/ domain". In Germany, a totally different in bachelor degree will definitely be rejected, for example, a student studying Business degree has no opportunity for accepting "Agriculture" or "Medicine" master's program, but could be accepted with bachelor in "Mangement", "Economics". The primary objective of this layer is to determine if a student's education background provides a sufficient theoretical foundation for a specific Master's program. Layer 2 addresses this by employing a "Reasoning" check. The system uses a lightweight LLM (Gemini 2.5 Flash) to evaluate if the student's major is a valid academic background for the program's requirements.

To transform qualitative reasoning into quantitative filtering, the LLM assigns a Match Score between 0.0 to 1.0 as defined in table below:

Table 6: LLM Score Rubric for Degree Compatibility

Score	Category	Description	Example
1.0	Exactly match or Synonym	The major is identical or a clear linguistic variation	Business Administration vs. Management
0.8 - 0.9	Strong Hybrid	Specific title where >70% of curriculum fits.	Digital Economy vs. Economics
0.6 - 0.7	Partial Match	The major contains the requirement as a	Media Design vs Communication Science

		component, but it is not the main focus	
0.3 - 0.5	Faculty Adjacent	The major is in the same general faculty, but the theoretical foundation is different.	Architecture vs. Civil Engineering
< 0.5	No Fit	The inferred curriculum has no meaningful overlap	Physics vs Economics

The implementation relies on Pydantic-based structured output to ensure the LLM returns valid JSON. This allows the system to programmatically filter out any program where the score falls below a threshold (0.5).

Role Prompting: The LLM is assigned the persona of a "Strict University Registrar."

Mitigating "Related Domain" Hallucination: One of the highest risks in automated admissions is the model hallucinating a connection between unrelated fields. To prevent this, the model is instructed that if a requirement contains vague terms like "related field" or "etc," it must not default to a "Pass". Instead, it must verify if the student's major belongs to the same General Faculty as the specific requirements. If the theoretical foundation (e.g., math depth or lab science) differs significantly, the model is directed to assign a low score (0.0–0.2) to prevent false positives.

Few-Shot Logic (Anchoring): The prompt provides explicit examples of what constitutes specific scores (e.g., explaining why "Mechatronics" vs. "CS" is a 0.6 while "Business Administration" vs. "Management" is a 1.0. This anchors the model's judgment and prevents score drift.

4.5.3 Layer 3: Semantic Ranking

The remaining candidates are technically eligible, but not necessarily well-aligned with the user's specific goals. Layer 3 ranks these candidates by semantic relevance.

The agent constructs a "User Persona" string combining the student's major, specific research interests, and career goals. It then generates vector embeddings for this persona and the program descriptions using GoogleGenerativeAIEmbeddings. A cosine similarity calculation determines the alignment between the user's intent and the curriculum content.

A key design challenge was balancing two conflicting search needs:

- Conceptual Understanding: If a student is interested in "Automotive Engineering", the system should also recommend "Vehicle Systems" or "Mobility Engineering", even if the words are different.

- **Specific Requirement Matching:** If a student specifically mentions a technical skill like "Python" or "Java", the system should prioritize programs that explicitly teach these tools over general programs that do not.

To resolve this, the system implements a Hybrid that runs two parallel scoring algorithms and fuses them using a weighted average.

1. The Semantic Pipeline

Technology: GoogleGenerativeAIEmbeddings (768-dimensional vectors).

Function: Converts the student's persona into a vector. It finds programs that are conceptually similar. It captures the user's intent, ensuring no relevant field is missed due to vocabulary differences.

2. Term Frequency-Inverse Document Frequency

Technology: TfidfVectorizer (Term Frequency-Inverse Document Frequency).

Function: Converts text into vectors that focus on distinct, rare keywords. If a student asks for a specific niche (e.g., "Electronic Commerce"), this pipeline heavily rewards programs containing that exact term. In another word, TF-IDF is an exact keyword counter, it doesn't care about the context, it only cares about rare, specific nouns.

The top candidates from this ranking are selected for the final score.

3. Final Score

The final score for Layer 3 is calculated using a weighted fusion of both pipelines. The system assigns a higher weight to the semantic signal to ensure the fundamental subject matter is correct, while using the lexical signal to refine the ranking.

$$S_{\text{Layer3}} = (S_{\text{semantic}} * 0.5) + (S_{\text{TF-RDF}} * 0.5)$$

S_{semantic} : Semantic Weight prioritizes broad conceptual fit.

$S_{\text{TF-RDF}}$: TF-RDF boosts programs that explicitly mention the student's specific interests.

By combining these two methodologies, the system minimizes the risk of Semantic Hallucination (recommending irrelevant fields) and Keyword Blindness (missing relevant synonyms). This results in a ranked list of candidates that are both conceptually aligned and technically specific to the user's needs.

4.5.4 Layer 4: ECTs Verification

Although in layer1, the system has already filter the required degree domains which plays as a first part to eliminate unrelated programs, the most important part is ECTs for each module requirement to verifying if the student's specific transcript meets the program's prerequisite module requirements.

The course names from student profile's PDF was collected and because of course names vary globally, exact keyword matching is insufficient. The *check_ects_match_with_embeddings* function employs a fuzzy matching logic. It generates embeddings for every course in the user's transcript. For each prerequisite domain (e.g., "Business Administration"), it calculates similarity scores against the transcript. Courses exceeding a similarity threshold of 0.5 are summed to check if they meet the credit requirement.

A significant challenge in ECTS verification is the semantic gap between high-level degree requirements and specific transcript course titles. For instance, a requirement labeled 'Business Administration' might reject a valid 'Marketing' course due to low cosine similarity.

To resolve this, the system implements a Semantic Enrichment step. Instead of embedding only the domain name, the system constructs a composite string that includes all listed sub-module. This technique expands the semantic footprint of the requirement vector. By explicitly including sub-fields (e.g., "Marketing", "Finance") in the embedding, the system significantly reduces false negatives for specialized courses that fall under broad administrative categories.

- Prevents the "Double Dipping" issue by subtracting credits from a student course once they are assigned to a requirement.
- Global Sorting: The Best Match is implemented first to avoid the biases result toward A if processing Requirements A, then B, then C in order. If a course matches Requirement B with 0.95 and Requirement A with 0.8, it will be assigned to B first.

The final score for ECTs verification is calculated by the percentage of meet requirement by total requirement.

For the programs which is not explicitly put the requirement for each module domain, the score for this layer is defined as 1.

For the student who did not state their PDF document, the score for it layer is 0.5.

4.5.5 Final Ranking and Output Generation

After the Deep ECTS Verification (Layer 4), every eligible candidate program possesses distinct scores for semantic fit, degree compatibility, and transcript coverage. To generate a single ranking metric, the system applies a weighted linear combination formula:

$$S_{\text{final}} = (S_{\text{degree compatibility}} * 0.1) + (S_{\text{semantic ranking}} * 0.5) + (S_{\text{ECTs}} * 0.4)$$

The weighting strategy prioritizes the user's research interests and the meeting requirements, ensuring the recommendations are highly relevant.

4.6 Agent 4: University Verifier

Agent 3 identifies programs that are eligible based on static attributes such as degree level and field of study. However, application success often depends on administrative requirements that vary by applicant nationality and are inconsistently documented across websites. Agent 4 addresses this limitation by interpreting university regulations in the context of the user's specific profile.

Agent 4 does not merely retrieve documents. Instead, it translates fragmented admission rules into a personalized and actionable checklist. To do so, it uses a Retrieval-Augmented Generation (RAG) pipeline constrained by explicit heuristic rules. This design connects generic university regulations to applicant specific administrative obligations.

4.6.1 Perplexity API

Identifying required application documents is not a simple lookup task. Universities typically distribute requirements across multiple sources, such as International Office pages, faculty websites, and formal examination regulations in PDF form.

Perplexity API was selected because it supports structured synthesis across heterogeneous sources. It offers three capabilities relevant to this task:

- **Autonomous Synthesis:** The system reads and integrates information from multiple authoritative pages rather than returning isolated links.
- **Noise Filtration:** The reasoning engine distinguishes between application and enrollment phases, which prevents contamination from post-admission requirements.
- **Source Grounding:** By grounding every extracted requirement in a verifiable live URL, it significantly reduces the hallucination risks inherent in static LLMs.

The process follows a strict Human-in-the-Loop (HITL) sequence:

1. The system presents the ranked list from Agent 3 to the user.
2. The user selects a subset of programs, typically three.
3. Agent 4 activates only after selection and performs live verification for each chosen program.

4.6.2 Reasoning Logic

The agent's behavior is constrained by explicit logical rules encoded in the prompt. These rules address recurring sources of ambiguity in university application requirements.

Country-Specific Requirements

The prompt enforces conditional logic based on citizenship:

- China, Vietnam, and India: If the applicant holds citizenship from one of these countries, the agent must check for an APS (Akademische Prüfstelle) requirement. If the university website does not mention APS explicitly, the agent must still output “APS Certificate (Standard Federal Requirement)”, reflecting its mandatory status for applicants who completed their bachelor’s degree domestically.
- Pakistan: The agent may check for HEC attestation but is explicitly forbidden from requesting APS, preventing false positives.

Temporal Filtering: Application vs. Enrollment

University websites frequently conflate documents required for application with those required after admission. To avoid misleading applicants, the agent applies a strict phase filter:

- Rule: Extract only documents required for the application phase. Exclude all enrollment-stage documents.
- Effect: Items such as health insurance proof or semester fee confirmation are removed from the checklist.

This ensures the output reflects requirements necessary to obtain an admission offer.

Semantic Standardization

Universities use inconsistent terminology for identical documents. The agent normalizes these terms using fixed vocabulary rules:

- “Bachelor Certificate” replaces variants such as “Proof of first degree.”
- “Transcript of Records” replaces phrases such as “Overview of grades.”

This normalization produces consistent outputs across institutions and programs.

Ambiguity Resolution: Mandatory vs. Optional Documents

Admissions pages often mix mandatory documents with recommended or advantageous materials. The agent parses linguistic qualifiers to distinguish between them:

- Documents labeled as “recommended,” “selection criteria,” or “bonus points” are marked as (Optional).
- All other documents are treated as mandatory unless explicitly stated otherwise.

This distinction helps applicants prioritize legally required documents.

Relevance Filtering: Logistics vs. Academic Criteria

University pages often include academic admission criteria (e.g., GPA thresholds, ECTS requirements) alongside document submission rules. The agent excludes such content:

- Included: Document format, translation requirements, notarization, and submission logistics.
- Excluded: Eligibility criteria and general program descriptions.

This keeps the output focused on actionable administrative steps.

4.7 Agent 5: Timeline Planner

Agent 4 produces a program-specific list of required application documents. However, meeting document requirements alone is insufficient if applicants fail to account for preparation time and external processing delays. Agent 5 addresses this issue by translating document requirements into a concrete preparation timeline anchored to each program's application deadline.

Agent 5 computes milestone dates by subtracting fixed offsets from the deadline. These offsets reflect known administrative bottlenecks in the German application process and are encoded as explicit scheduling rules.

The scheduling logic is governed by four primary rules:

Country-Specific Requirements

- If Agent 4 identifies an APS certificate requirement, Agent 5 schedules the start of the APS process at deadline – 180 days.
- APS certification is the longest and least flexible bottleneck in German university admissions. The process involves document pre-screening, limited interview availability, and post-interview evaluation, which together can span several months. The six-month offset reflects conservative preparation time and ensures applicants recognize this constraint early.

Uni-Assist Processing Buffer

- If the application route is Uni-Assist or Preliminary Review Documentation (VPD), Agent 5 subtracts a 30-day buffer (deadline – 30 days).
- Uni-Assist typically requires four to six weeks to verify submitted documents. Submissions made close to the official deadline risk missing verification cutoffs, which can delay or invalidate applications.

Translation Milestone

- Agent 5 schedules document translation at deadline – 60 days.
- Most universities require certified translations into German or English. This offset prompts early coordination with translators and notary services, reducing last-minute delays.

Conflict Resolution (Triage Logic)

- If a computed start date lies in the past relative to the current date, Agent 5 triggers a conflict resolution routine.
- The agent replaces the calculated date with the current date and labels the task as Overdue/Urgent. This prevents the display of infeasible timelines and explicitly signals schedule risk to the user.

Agent 5 returns a consolidated timeline for each selected program, enabling applicants to assess feasibility, prioritize tasks, and make informed decisions about whether to proceed with an application.

4.8 Agent 6: PDF Report Generation

Agent 6 is the final stage of the workflow. Its function is to consolidate the intermediate outputs produced by Agents 1 through 5 into a persistent artifact: the Application Strategy Dossier in PDF format.

4.8.1 Input and Output

Agent 6 follows a strict state-to-file design. It consumes a unified JSON object that aggregates three data streams:

- User context (Agent 1): Static applicant attributes such as nationality, academic background, and target field.
- Verified logic (Agents 4 and 5): Derived administrative requirements and timeline constraints, including APS flags and application deadlines.
- Program metadata (Agent 3): Program match scores and explanatory reasoning.

The agent produces a multi-page PDF titled Application Strategy Dossier. The document presents the information using structured tables, section headers, and visual alerts to highlight deadlines and critical requirements.

4.8.2 Purpose of the PDF Strategy Dossier

The workflow concludes with a static PDF rather than a dynamic dashboard or chat-based summary for three reasons.

- Persistence: German university applications often span several months. Chat-based outputs are transient, whereas a PDF provides a durable reference that users can store, print, and revisit throughout the process.
- Consolidation: Information is generated incrementally across agents, each with a narrow responsibility. The PDF integrates applicant profile data, document requirements, and

timelines into a single coherent view, making explicit why each recommendation applies to the user.

- Usability: A structured document supports planning and task tracking more effectively than unformatted text. By presenting requirements and deadlines in a consistent layout, the dossier reduces cognitive load during long-term application preparation.

5 Evaluation

This framework evaluates a 6-agent system that matches international students with German Master's programs using synthetic test profiles only.

Overview of Architecture:

- Agent 1: User Profile Extraction & ECTS Conversion.
- Agent 3: 4-Layer Filtering & Ranking (Hard Constraints - Degree Compatibility - Semantic Matching - ECTS Validation).
- Agent 4: Document Checklist Generation.
- Agent 5: Application Timeline Planning.
- Agent 6: PDF Report Generation.

The evaluation of this system will be included Two Level Testing:

- Component-Level Tests: Validate each agent and core function separately with 3 synthetic profiles.
- End-to-End Tests: Validate complete workflow integration.

5 synthetic student profiles is applied to across all tests. Profiles are designed to cover diverse backgrounds: Vietnam, India , Spain, China and Germany.

5.1 Profiles Design

Profile 1: Standard Applicant (Vietnam)

ID: PROFILE-001-VN-CS

Background:

- Origin: Vietnam (APS Required).
- Academics: B.Sc. Computer Science, GPA 3.5/4.0.
- Core Constraint: Strict budget (< €3000/semester), Winter Semester, Prefer Cities (Munich, Stuttgart, Karlsruhe), Desire's Program (Data Science, Artificial Intelligence).

Design Purpose:

- Test Administrative Logic (Agent 4 & 5): This profile serves as the primary validation for the APS Logic. Success is defined by the system correctly identifying the mandatory "APS Certificate" requirement and applying a 6-month buffer to the application timeline.
- Test Academic Matching (Agent 3): Validates the system's ability to identify "Direct Match" programs where the Bachelor's Degree perfectly aligns with the Master's.

Profile 2: Business Administration Applicant (India)

ID: PROFILE-002-IN-BUS

Background:

- Origin: India (APS Required).
- Academics: B.A. Business Administration, GPA 75/100.
- Constraint: Strict budget (< €5000/semester), Desire's Program (Business Management), Summer Semester

Design Purpose:

- Test Semantic Mapping: This tests Agent 3 (Matcher)'s ability to map a general "Business" degree to specialized management programs (e.g., "Digital Business Management") rather than irrelevant technical degrees.
- Test Temporal Filtering: Validates if the system correctly filters for "Summer Semester" intakes, which are significantly rarer than Winter intakes.

Profile 3: Economics Applicant (Spain)

ID: PROFILE-003-ES-ECON

Background:

- Origin: Spain (EU Citizen).
- Academics: B.Sc. Economics, GPA 6.5/10.
- Constraint: High Budget (€10,000), Flexible.

Design Purpose:

- Test Negative Logic: This profile acts as a control. The success condition is the absence of warnings. The system must not trigger APS checks, APS timelines, or non-EU tuition fees.
- Test Fee Logic (Agent 6): Validates if the Reporter correctly identifies "EU Tuition: €0.0" instead of the Non-EU fees applied to the other profiles.

Profile 4: Domestic Applicant (Germany)

ID: PROFILE-004-DE-WINF

Background:

- Origin: Germany (Domestic/EU)

- Academics: B.Sc. Business Informatics (Wirtschaftsinformatik), Grade 2.3 (German Scale), 210 ECTS.
- Constraint: Budget (< €2000/semester), Winter Semester, Prefer Cities (Stuttgart, Mannheim), Desire's Program (Business Informatics, Digital Business)

Design Purpose:

- Test "Native" Data Handling (Agent 1): Validates that the system correctly identifies a domestic applicant and skips the grade conversion step (Agent 1 should recognize "2.3" is already in the target format).
- Test Fee Logic (Agent 6): Validates if the Reporter correctly identifies "EU Tuition: €0.0" instead of the Non-EU fees applied to the other profiles.

Profile 3: Marketing Applicant (Pakistan)

ID: PROFILE-005-PK-MKT

Background:

- Origin: Pakistan (Non-EU, No APS)
- Academics: B.Sc. Marketing, GPA 3.2/4.0
- Constraint: Budget (< €4000/semester), Flexible Semester (No Preference), Work Experience (24 Months), Desire's Program (Marketing, Digital Marketing)

Design Purpose:

- Test Geo-Political Granularity (Agent 4): This is a critical "False Positive" test. While Pakistan is geographically close to India (an APS country), it does not require an APS certificate. The system must apply Non-EU visa logic without triggering the APS warning.
- Test "Next Available" Logic (Agent 5): Since the semester preference is "None," Agent 5 must dynamically calculate the nearest upcoming intake (Winter or Summer) rather than filtering for a specific season.

The Detail synthetic Profile is attached in the Appendix A.

5.2 Test Sample Database:

Distribution of the 20 master's programs: this test database includes

Geographic Distribution

- State: All 20 programs are from Baden-Württemberg
- Cities:

- Stuttgart: 2 programs
- Heidelberg: 2 programs
- Heilbronn: 3 programs
- Tübingen: 3 programs
- Ulm: 2 programs

Application Mode

- Direct: 17 programs (85%)
- VPD: 3 programs (15%)

Tuition Fees (per semester)

- Free (€0): 17 programs
- Paid: 3 programs
- €750: 1 program
- €4,500-€5,040: 2 programs (SRH University - private)
- €7,300: 1 program (Zeppelin University - private)

Non-EU Tuition Fees

- €1,500: 18 programs (standard Baden-Württemberg fee)
- €5,450-€6,950: 2 programs (private universities)

Field Distribution

- Computer Science/AI: 5 programs (25%)
- Engineering: 3 programs (15%)
- Electrical Engineering, Mechatronic Systems, Chemical Engineering
- Economics/Business: 4 programs (20%)
- Economics, Economics & Finance (2), Corporate Management & Economics, International Business
- Life Sciences/Biotechnology: 5 programs (25%)
- Bioinformatics, Chemistry, Biotechnology, Life Science Innovation, Food Science
- Agriculture: 1 program (5%)
- Landscape Architecture: 1 program (5%)
- Dance Movement Therapy: 1 program (5%)

GPA Requirements (German scale)

- No minimum specified: 14 programs (70%)
- 2.5 (good): 5 programs (25%)
- 3.0 (satisfactory): 1 program (5%)

English Language Requirements

- B2 level: 19 programs (95%)
- C1 level: 1 program (5%)

Semester Availability

- Winter semester only: 9 programs (45%)
- Both semesters: 10 programs (50%)
- Summer semester only: 1 program (5%)

Work Experience & Internship

- No work experience required: All 20 programs (100%)
- Internship required: 1 program (Landscape Architecture)

ECTS Requirements

- No specific ECTS requirements: 11 programs (55%)
- With specific ECTS requirements: 9 programs (45%)
- Most common: Economics/quantitative methods requirements

5.3 Component-Level Evaluation

5.3.1 Agent 1 Evaluation

Agent include 4 functionalities:

1. PDF Parsing: Extracts the bachelor course's name and credits of each course.
2. Profile Field Extraction: Extracts structured data from unstructured natural language input.
3. ECTS Conversion: Converts various credit systems to ECTS and ECTs factor.
4. GPA Conversion: Converts various GPA scales to the German grading system.

Table 7: Testing Data Profile

Profile ID	Citizenship	Bachelor Field	Credits	Semesters	GPA Scale
VN-CS	Vietnam	Computer Science	130 CP	8	4.0
IN-BUS	India	Business Administration	180 CP	6	100
ES-ECON	Spain (EU)	Economics	240 ECTS	8	10.0

DE-BINF	Germany	Business Informatics	210 ECTS	7	5.0
PK-MKT	Pakistan	Digital Marketing	128 CP	8	4.0

For each profile, expected values were manually calculated:

- Expected ECTS: Calculated using the ECTS formula
- Expected German GPA: Converted using the Bavarian formula
- Expected Fields: All mandatory fields defined in the system schema

Evaluation Metrics:

The performance of Agent 1 was assessed against three defined targets:

Test C1-01 (ECTS Accuracy): 100% accuracy

Test C1-02 (Completeness): 90% accuracy

Test C1-03 (GPA Precision): 100% accuracy

These targets settled 100% for ECTS and GPA because they are the important factor and the system run base on the logic predefined formula. Test C1-02 is settled 90% because the system run base on the prompt AI model for extraction the natural language from student. Small errors could happen.

Results:

Test C1-01

This unittest will test the ECTS converted from Credit Points System of Student's Bachelor Degree.

Table 8: Test C1-01 (ECTS Accuracy)

Profile ID	Original Credits	Semesters	Expected ECTS factor	Calculated ECTS factor	Error	Status
VN-CS	130	8	1.85	1.85	0.0	PASS
IN-BUS	180	6	1.0	1.0	0.0	PASS
ES-ECON	240	8	1.0	1.0	0.0	PASS
DE-BINF	240	7	1.0	1.0	0.0	PASS

PK-MKT	128	8	1.88	1.88	0.0	PASS
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Test C1-02

This unittest will test the ability to extract important/ necessary information from student's input (natural language).

Expected field's extraction including:

Extract Full Name

1. Extract Full Name
2. Extract Citizenship
3. Extract Bachelor Field
4. Extract Total Credits
5. Extract Semesters
6. Extract GPA Score
7. Extract GPA Max Scale
8. Extract GPA Min Score
9. Extract Interests Coverage
10. Extract English Test
11. Extract English Score
12. Extract Work Experience Months
13. Extract Max Tuition
14. Extract Preferred Semester
15. Extract Preferred Cities
16. Extract Preferred State

Table 9: Test C1-02 (Completeness)

Profile ID	Total Fields	Extracted Fields	Error / Missing Fields	Completeness %	Status
VN-CS	16	15	1	94%	PASS
IN-BUS	16	16	0	100%	PASS
ES-ECON	16	16	0	100%	PASS
DE-BINF	16	16	0	100%	PASS
PK-MKT	16	16	0	100%	PASS

The error field happened in VN-CS profile due to the working experience. This is one time input test, it could be improve through the end to end test when directly interact with Agent1.

Natural language input	Expected extraction	Result
I have no working experience	[]	Null

Test C1-03

This unittest wil test the correct German GPA converted from the GPA system from student's Bachelor Degree.

Table 10: Test C1-03 (GPA Precision):

Profile ID	Original GPA	GPA Scale	Min Passing	Expected German GPA	Calculated German GPA	Status
VN-CS	3.5	4.0	2.0	1.75	1.75	PASS
IN-BUS	75	100	40	2.25	2.25	PASS
ES-ECON	6.5	10.0	5.0	3.10	3.10	PASS
DE-BINF	2.3	1.0	5.0	2.3	2.3	PASS
PK-MKT	3.2	4.0	2.0	2.0	2.0	PASS

The agent 1 achieved 100% accuracy with zero deviation, surpassing the 100% target

- Perfect ECTS Conversion: The ECTS conversion formula performed flawlessly across all 3 profiles with zero errors.
- Perfect Field Extraction Rate: 98.8% completeness demonstrates robust natural language understanding.
- Accurate GPA Conversion: 100% accuracy across three different GPA scales (4.0, 10.0, 100)
- Mandatory Field Reliability: 100% extraction rate for all mandatory fields ensures downstream agents receive complete data.

5.3.2 Agent 3 Evaluation

Agent 3 has been tested with 5 diverse test profiles against a sample database of 20 master's programs. The testing framework evaluates Agent3's ability to:

- Layer 1: Hard Constraint matches student data against program requirements (GPA, Tuition, City, English Level, Work Experiencp, Semester, and ECTS credits).
- Layer 2: Degree Compatibility (Semantic Match) Evaluates how well the student's Bachelor's field aligns with the target Master's domain (categorized as HIGH, MEDIUM, LOW, or NO).
- Layer 3: Relevance choose based on the alignment between student interests and specific course content summaries.
- Layer 4: Final Ranking (Decision) Check ECTS and give the final selection of the programs recommended to the user.

Evaluation Metrics:

To ensure the AI's reliability, a comparison is made from the Agent's output against a Manual Excel file

- Metric: Accuracy (Total matches between Agent and Manual / Total profiles).
- Threshold: >90% Agreement.
- Process: Each layer was tested independently to identify errors might occur (e.g., if Layer 1 passes a student but the Manual check says NO).

The test is runned one by one profile through each layer, here is the testing result followed by each profile:

5 profiles x 20 programs = 100 test cases

Table 11: Test C1-04: 100 test cases for 4 layers of Agent3

Layer	Filter Type	Expected Pass Pro-gram	Actual Pass Programs	Percentage Passed	Status
1	Hard constraints	47	47	100%	PASS
2	Degree Com- patibility	24	26	92%	PASS
3	Semantic Matching	4	4	100%	PASS
4	ECTS Verifi- cation	4	4	100%	PASS

To verify the effectiveness of the 4-layer architecture, we benchmarked Agent3 against a ground-truth dataset. In a direct comparison, the 4 programs identified by manual audit were perfectly matched by the agent's output. Furthermore, a broader evaluation of 100 test cases yielded a 90% consistency rate between the agent's automated filtering and manual checking. This indicates that Agent3 effectively replicates the complex logic required for student-program matching with a high degree of precision.

Since Agent4, Agent 5 and Agent 6 will extract information from direct university webpage, calculate a reasonable timeline and extract PDF planing respectively, the test for accuracy and success extraction will be evaluate for full profile in below section.

5.4 End-to-end Evaluation

Scope of TestingThe:

The final evaluation stage involved executing the full agentic pipeline against the complete production database (106 Programs). Unlike the component tests which compared results against a small, manually labeled "Gold Standard," this End-to-End test focused on System Validity and Constraint Adherence. The objective was to verify that the final PDF artifact contains logically consistent, valid recommendations that strictly adhere to the user's hard constraints.

Integration test involves processing 5 diverse student profiles against Master's programs within the state of Baden-Württemberg. For this validation phase, the system was configured to output a maximum of N=2 top-ranked programs per student to test the agent's ability to prioritize the best matches.

Experimental Setup Input:

- Synthetic Test Profiles.
- Dataset: Full Production Database (N=106 Programs).
- Output: 5 Generated "Strategy Dossier" PDFs (extracting Top 2 recommendations per profile).
- Method: Manual Forensic Audit of the generated PDF content.

Evaluation Metric: PDF Dossier vs. Official Website Accuracy.

To assess the end-to-end accuracy of the generated "Application Strategy Dossier" (as a PDF file), each output was audited against the official university website using five distinct Verification Checks.

1. Check 1: Hard Constraint Adherence

- Objective: Verify that the AI never recommends a program that violates the user's non-negotiable limitations.Metrics:Budget

- Compliance: The recommended Master's degree field must be a valid progression from the user's profile.
- Success Condition: 100% Pass Rate. Any recommendation violating these limits is considered a Failure.

2. Check 2: Specific Country Requirement

- Objective: Ensure the system correctly identifies and warns about country-specific administrative hurdles.
- Metrics: APS Detection: For applicants from China, Vietnam, or India, the "APS Certificate" must be listed as a Mandatory Document.
- Success Condition: The "APS Warning" block appears if and only if the citizenship requires it.

3. Check 3: Financial Verification

- Objective: Confirm that the system applies the correct tuition fee structure based on the user's legal status.
- Metrics: Citizenship-Based Pricing: If Non-EU: Display "International Tuition" (e.g., €1,500/semester in BW). If EU/EEA: Display "Domestic Tuition" (usually €0.00 or nominal fees).
- Success Condition: The displayed fee in the PDF matches the official fee regulation for that specific citizenship.

4. Check 4: Accuracy

- Objective: Validate that application deadlines are precise and buffers are applied correctly.
- Metrics: Deadline Precision: The specific date (e.g., "July 15") matches the current academic year's official deadline on the university website.
- Buffer Logic: Standard: Deadline shown as is a High-Risk: A "Start Date" buffer of -180 days is applied for APS candidates.
- Success Condition: The timeline is factually correct (date matches website) and strategically safe (includes necessary buffers).

5. Check 5: Information Extraction

- Objective: Measure how many required documents the AI successfully found compared to the full official list.
- Metric: The percentage of total extracted document divided by total required document
- Success Condition: The agent must find all critical docs like Transcripts/Degrees, even if it misses minor formatting notes.

Integrated Performance Results

Data Extraction Reliability: Evaluation of the agent's ability to pull correct details from the 106-program pool.

Workflow Consistency: Confirming that the automated pipeline produces a professional PDF Dossier without manual formatting.

The test is runned one by one profile through the whole system, here is the testing result followed by each profile:

5 profiles x top 2 selected programs = 8 test cases

The reason that VN-CS and DE-BINF only have 1 matching program is the constraint based on preferences.

Results:

Table 12: Result of End-to-end Testing

Profile ID	Programs	Check 1: Hard Constraint	Check 2: Specific Country Requirement	Check 3: Financial Verification	Check 4: Accuracy	Check 5: Information Extraction (document checklist)
VN-CS	Computational Linguistics - University of Stuttgart	PASS	PASS (APS Warn)	PASS	PASS	2/5 FAIL
IN-BUS	International Management (MBA) - Nürtingen-Geislingen University	PASS	PASS (APS Warn)	PASS	PASS	7/7 PASS
IN-BUS	Master of Science in Digital Business Management - Nürtingen-Geislingen University	PASS	PASS (APS Warn)	PASS	PASS	4/4 PASS
ES-ECON	Master of Science in	PASS	PASS (No Warn)	PASS	PASS	4/4 PASS

	International Business and Economics (IBE) - University of Hohenheim					
ES-ECON	International Finance - Nürtingen-Geislingen University	PASS	PASS (No Warn)	PASS	PASS	6/6 PASS
DE-BINF	Master of Science in International Business and Economics (IBE) - University: University of Hohenheim	PASS	PASS (No Warn)	PASS	PASS	5/5 PASS
PK-MKT	Management & Digital Technology (MSc) - Technical University of Munich	PASS	PASS (HEC Attestation Note)	PASS	PASS	6/6 PASS
PK-MKT	Master of Arts in Marketing and Artificial Intelligence - Aalen University	PASS	PASS (HEC Attestation Note)	PASS	PASS	4/6 FAIL

The audit of 10 program recommendations across 5 distinct user profiles demonstrates that the system achieved 100% Critical Safety Accuracy, and 6/8 correct extracted document Information.

1. Safety and Constraint Logic (100% Reliability)The system demonstrated flawless execution of all "Gatekeeper" logic.Geo-Political Adaptation: In every applicable case (P1-Vietnam, P2-India), the system correctly inserted the APS Warning Block and shifted the timeline by -180

days. Conversely, for P4-Germany, these warnings were successfully suppressed, proving the dynamic nature of the report generation and for P5-Pakistan, HEC Stamp is added in the end of PDF as a Note.

2. Fiscal Accuracy: The system correctly differentiated between EU (€0) and Non-EU tuition fees in all cases, ensuring no applicant received misleading financial data.
3. Hard Constraints: No program violating the user's budget or degree requirements was recommended.
4. Information Extraction and Limitations: While the system successfully extracted all mandatory documents (Transcripts, Degrees, Language Certificates), it achieved a slightly lower recall score in 2 out of 8 cases.
5. Identified Failure Mode: The missing data points were exclusively related to formatting nuances rather than core requirements. Example: For P1-B (Uni Stuttgart) and P5-B (Uni Aalen), the agent missed a requirement (CV, VPD, University entrance) because this specific instruction was hosted on a sub-link (e.g., a PDF guideline) rather than the main admission page. Cause: The current prompt architecture focuses on the main text of the retrieved URL. If a university fragments its requirements across multiple sub-pages or downloadable PDFs, the agent may miss secondary formatting details.

Conclusion: The system is successfully extract 5 PDF Dossier for 5 Profile. The evaluation confirms that the system is Safe, Valid, and Highly Accurate. The minor recall gaps regarding formatting do not compromise the applicant's eligibility, as the core legal and academic requirements were captured with 100% precision. While the system demonstrated accuracy in enforcing critical safety logic and applicant eligibility constraints, the granular document retrieval process revealed limitations in extracting formatting nuances from nested sub-pages, a challenge detailed further in Chapter 6: Discussion and Future Work.

6 Conclusions

6.1 Summary of Findings

This research demonstrated the efficiency of an Agentic AI architecture in automating the German university application process for international students. Through the testing of five distinct applicant profiles against a control database, the system validated four core capabilities.

Automated Strategy Dossier Generation: The system successfully consolidated complex administrative logic into a tangible, human-readable artifact. For every test profile, the output agent generated a personalized "Application Strategy Dossier" (as a PDF file) containing a program summary, a tailored document checklist, tuition breakdown, and step-by-step timeline. This confirms the system's ability to providing users with a downloadable roadmap that persists beyond the chat session.

High Precision Top-N Recommendation: The evaluation validated the ranking engine's ability to distill a database of 106 programs down to a precise Top-N list that strictly adheres to user preferences. The system achieved a successfully balancing "Hard Constraints" (Budget, Degree Eligibility, ..) with "Soft Preferences" (Interests, Preferred Cities, ..). This proves that the weighted scoring algorithm effectively quantifies holistic fit, ensuring users are presented only with viable, high-quality options

Geo-Political Granularity: The system successfully distinguished additional requirements for Non-EU applications. First, the system proved highly effective at enforcing bureaucratic and legal constraints. The evaluation confirmed accuracy in applying geo-political safety logic. Applicants from high-risk origins, specifically Vietnam, China, Mongolia and India, automatically received correct timeline adjustments with a 180-day buffer for the APS certificate. Conversely, candidates from the European Union received standard timelines, validating the system's ability to dynamically adapt to citizenship rules without manual intervention.

Personalization: The reporting agent demonstrated accuracy in adapting financial data to the user's legal status. The system dynamically adjusted the reported tuition fees based on citizenship. For example, Spanish applicant (Profile 4), the dossier correctly listed the tuition as 0.00 EUR, reflecting EU treaty rights. In contrast, for the Pakistan, India, and Vietnam profiles applying to the same universities in Baden-Württemberg, the system automatically displayed the mandatory 1,500 EUR

Non-EU semester fee. This confirms the system's ability to provide legally accurate financial projections rather than static database values.

6.2 Limitations

Transnational Applicants : The current system operates on the assumption presumes the applicant's country of citizenship aligns with the country where they obtained their Bachelor's degree. In complex scenarios, such as an Indian citizen obtaining a degree in the United Kingdom, the system prioritizes the citizenship rule (triggering APS requirements) while potentially misinterpreting the academic grading scale (using Indian conversion logic instead of UK logic). This simplification was necessary for the prototype but restricts accuracy for Transnational Applicants who possess credentials from jurisdictions different than their passport origin.

Degree Compatibility and ECTS: While the system performs title-level matching for degree compatibility, it cannot guarantee 100% accuracy in ECTS validation. In official admissions, university registrars manually compare detailed "Module Descriptions" (Modulhandbücher), often 50+ page PDFs to verify that specific learning outcomes match. Due to the high computational resource cost and retrieval latency required to parse and analyze thousands of syllabus pages, the current system relies on semantic title matching. Consequently, it may generate false positives where a course title appears relevant but lacks the specific theoretical depth required by the target university.

Conditional Admissibility: The logical model treats admission requirements as binary constraints (Pass/Fail). It currently lacks the nuance to handle Conditional Admission rules often found in German universities. For instance, if an applicant misses a specific subject requirement (e.g., "Missing 5 ECTS in Statistics"), universities often grant admission on the condition that the student can complete later. The current system flags these profiles as immediate mismatches, thereby potentially excluding viable candidates who could satisfy these conditions post-admission.

Hierarchical Information Retrieval Gap: As evidenced in the evaluation, the system struggles with Deep Web information architectures. In cases where critical formatting requirements were hosted on nested sub-pages or external PDF downloads, the agent may fail to retrieve them. The current single-layer scraping architecture is insufficient for fragmented university websites where requirements are decentralized across multiple sub-domains.

Data Privacy and GDPR Compliance: The processing of academic transcripts and personal constraints involves Sensitive Personal Identifiable Information (PII). While the current prototype functions in a controlled test environment, it does not yet implement the full "Right to Erasure" or data anonymization protocols required by the General Data Protection Regulation (GDPR) for a public deployment. This limits the system's immediate viability as a commercial product without further architectural changes to ensure strict data sovereignty.

6.3 Future Work

Deployment of a User-Centric Interface (UI): While the current prototype relies on a functional testing interface, future development will focus on launching a Production-Grade Web Application. This interface will transition from a linear chat stream to a Dynamic Dashboard, allowing users to:

- Visualize Timelines: Interactively adjust start dates to see how deadlines shift in real-time.
- Edit Constraints: Manually override extracted profile data (e.g., GPA or Budget) without restarting the chat session.
- Preview Deliverables: View and download the PDF file directly within the platform. This shift to a robust UI is essential to improve user trust and reduce the perceived latency of the agentic workflow.

Integration of Open-Ended Conversational Capabilities: While the current agentic workflow excels at structured task execution, it remains rigidly bound to the application strategy pipeline. Future iterations should evolve into a Hybrid Architecture by integrating a "General Inquiry Agent" powered by Retrieval-Augmented Generation (RAG). This would allow the system to step outside the structured workflow to answer ad-hoc, open-ended questions from applicants (e.g., "What is the average cost of living in Munich?" or "How do I open a German blocked bank account?"). By combining highly structured logical reasoning with general conversational flexibility, the system can serve not only as a strategic planner but as an all-encompassing knowledge base, further centralizing the applicant's research process.

Dual-Sided Deployment: While this thesis focused on the applicant's perspective, the core architecture, specifically the Degree Checking and Hard Constraint agents, holds significant potential for university administration. Future development could adapt the system into a "Pre-Screening " for admission offices. Instead of students querying for programs, universities could feed thousands of incoming PDF applications into the system. The agent would then automatically flag ineligible candidates before a human officer ever reviews the file, drastically reducing the administrative burden of processing high-volume international applications. However, to render this "reverse use case" viable and GDPR-compliant, the architecture must evolve into a Local LLM implementation.

By deploying open-source models directly on on-premise university servers, the system ensures that sensitive applicant data never leaves the institution's secure infrastructure, enabling high-volume automation without compromising data sovereignty.

Recursive Deep Web Crawling: To address the retrieval limitations identified in the evaluation, the scraping architecture must evolve from a single-pass reader to a Recursive Agent. This agent would be empowered to identify and traverse nested sub-links (e.g., "Download Admission Requirement" or "View Examination Regulations") to index auxiliary documents. This upgrade is necessary to achieve near-perfect recall for formatting nuances.

6.4 Conclusions

This thesis confirms that Large Language Models, when orchestrated in a modular agentic workflow, can effectively democratize access to complex educational systems. The primary finding of this research is the system's ability to transition the user experience from passive information search to active Strategy Generation.

This system functions as a comprehensive consultant. It successfully consolidates complex regulations into a legally accurate Application Strategy Dossier (as a PDF file). This tangible artifact provides international applicants with a clear, step-by-step roadmap, reducing the cognitive load required to navigate foreign administrative procedures.

Furthermore, the system demonstrated high precision in generating Top-N Recommendations. By balancing Hard Constraints with Soft Preferences, the system ensures that users are presented only with viable, high-quality options. This was validated by the system's ability to dynamically calculate real-time deadlines and accurate tuition fees based on the user's specific citizenship, preventing the critical errors common in static database searches.

Finally, these findings suggest a significant shift in how academic admissions can be managed. The same logic that helps a student find the right program can be reversed to help a university identify the right student. By automating repetitive tasks, such as checking deadlines and verifying eligibility constraints while strictly adhering to data privacy standards, the system allows both applicants and admission officers to focus on what matters most: the qualitative potential of the candidate

Appendix A: Interview's result from Experts

Expert Roles in a University:

Expert A: Head of Faculty Department

Expert B: Head of Faculty Department

Expert C: Research Assistant (joint interview with Expert B)

Expert D: International Office Representative

Expert E: International Coordinator

Expert F: Student Center (Studi Center) Representative

Questions and Result:

Question 1: In your advising role, could you describe the balance between common, repetitive questions from applicants versus complex, unique cases that require your personal expertise?

- Expert A: In my advising role, at least 80-90% of the questions are general and very repetitive. Most of these inquiries are already covered in our prepared list for Q&A sessions. The most frequent questions are "Am I eligible based on my graduate Bachelor program?" and "What documents are requested?". We see a clear split: only 5% of questions come from German students, while a huge amount comes from international students.
- Expert B & C: The most frequent question we receive is "Am I eligible?". The quality of these inquiries varies greatly; some provide no information, others send frustrated emails attached with different PDFs. Based on the volume, 90% comes from international students, especially from non-European countries. It is a huge workload to check each application individually.
- Expert D: We do not have an exact number because we are required to check all applications. Our role is to check if the formal requirements are completely fulfilled. We calculate the GPA and then pass the program to the faculty. The biggest reason for rejection at our stage is if the institution is not recognized which student can check by themselves through Anabin; the final rejection is done by the faculty.
- Expert E: During the application period, my role acts as a consultant for students. I would say 99% to 100% of the inquiries are common questions, primarily "Am I eligible for this program?". The answers to these questions are usually already available on our website, so my response mostly consists of providing the specific link. When students ask if their specific Bachelor's degree makes them eligible, they could actually check this themselves through DAAD, but they often ask us instead.

- Expert F: From the Student Center's insight, the volume is actually quite balanced between domestic and international students. However, we frequently encounter the "Am I eligible?" question. Since our task covers the general university administration side, we often have to forward these specific academic queries to the responsible faculty departments or International Office.

Question 2: Beyond your own university website, what are the most important external platforms (like DAAD, mygermanuniversity, etc.) where your program information must be accurate? How do you manage updates across these channels?

- Expert A: DAAD has a very good process. They offer a free account, but also a paid account which allows for more details, stories, and images. We update this information twice a year; a coordinator reviews the content and uploads it online if there is a change.
- Expert B & C: External platforms are mostly handled by the International Office. The marketing department provides information before the application phase or combines efforts to integrate with the International Office. But DAAD also a reliable website from German academic exchange service for international students
- Expert D: We use Anabin to check university recognition and our own in house system to check data ourselves. For suspicious rarely cases, we send it to external institution for thoroughly checking For students, MyGermanUniversity and DAAD can be reliable sources.
- Expert E: Our university cooperates with MyGermanUniversity and DAAD. For DAAD, we update the information once or twice each year. The process is established: when I receive a change request from the faculty, I send it to our specific contact person at DAAD, and they make it go live. Generally, the requirements do not change frequently; we usually review them once a year.
- Expert F: Expert indicated that managing external platform updates is not relevant to their specific role at the Student Center.

Question 3: From your experience, what are the most common and critical mistakes international applicants make that lead to an incomplete or rejected application?

- Expert A: Common mistakes include missing documents (such as forgetting to upload the Transcript of Records). Applicants often do not know about the APS or specific requirements for countries like India and Pakistan. Furthermore, many do not fix their requirements regarding background or eligible grades. Approximately 30% are rejected because they do not fulfill formal eligibility requirements.
- Expert B & C: We see many missing documents and specifically missing APS certificates from certain countries. For Pakistan, the stamp is mandatory, yet they frequently forget it.

Applications are often not structured or well-prepared. Recently, we noticed applicants lean on ChatGPT to write their applications but then fail the interview. They often expect a clearer webpage but do not prepare their own documents well.

- Expert D: The main issues are missing documents and students not reading the requirements for international students carefully. A frequent error is the wrong GPA conversion, which is why we want to do a pre-check.
- Expert E: The most frequent issues are missing documents and students continuing to ask about eligibility when they could verify it themselves. A specific academic problem arises with Modules and ECTS; students need to check module requirements, but I often find they are confused between the specific module content and the general course names.
- Expert F: We commonly see missing documents. Furthermore, applicants often do not know about the additional requirements specific to the program or the university beyond the basic checklist.

Question 4: Thinking about an AI assistant that could manage a student's entire application (matching, summarizing requirements, building timelines), what is your single biggest hope and your single biggest concern with such a tool?

- Expert A: It would be great, especially alongside DAAD, as we do not really have such a tool. My hope as a university representative is to help reduce the huge workload and receive fewer emails with general questions (answering that 80-90% amount). It would be helpful for applicants to see if they have "no chance" regarding minimum requirements before applying.
- Expert B & C: A good review of what programs exist and new options would be valuable. Crucially, the system should match the content of each module. This is important because the name of a Bachelor's degree may differ between countries, but the field is not totally different
- Expert D: It helps us a lot if it is more transparent for the student. The International Office would welcome such help. There is already an approach called the PIM project, which researches AI to check credits and module descriptions; this integrates with many universities already and is useful for exchange students.
- Expert E: My hope is to reduce time and workload, especially for the faculty programs. This would allow them to concentrate on students who really meet the requirements without having to scan through applications with missing documents. For the students, it would save time spent waiting for answers, which is particularly helpful given the different time zones. It could open a lot of opportunities for students by streamlining the process.

- Expert F: hope it can reduce the general workload. Ideally, the system should answer not only application requirements but also surrounding problems, such as questions about location and visas. I also see significant value in saving the waiting time for responses.

Question 5: From a university's perspective, what are the most important factors required for you to trust and perhaps even recommend an AI tool like this to your potential applicants?

- Expert A: Accuracy is exactly what is important. Beside that, it would be great to have such an AI system; I do not have any concerns.
- Expert B & C: We would really like to have an in-house system when working with profiles, but it must strictly follow GDPR and be transparent. There are concerns regarding IT security if students upload their entire CV online regarding German law.
- Expert D: We have an open opinion but need technical support, as we are concerned about software "doing all the things." At the moment, I cannot see 100% trust, but perhaps in the future
- Expert E: My only concern is that personal information should be secure. Apart from security, I do not have any other concerns. I would definitely use such an AI system to support my work and would recommend it to students if it is properly evaluated.
- Expert F: I would say I do not really trust it yet because of accuracy; that is my biggest concern. However, if the system is evaluated, trusted by many students, and proven helpful, I would definitely recommend it.

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Overview of AI-Supported Tools Used

In accordance with the declaration of authenticity regarding the use of artificial intelligence, the following tools were utilized during the creation of this thesis and its accompanying software implementation:

1. ChatGPT (OpenAI)

- Product Name: ChatGPT
- Source (URL): <https://chatgpt.com/>
- Software Functions Used: Preliminary brainstorming, topic exploration, and initial concept generation.
- Extent of Use: ChatGPT was utilized at the very beginning of the thesis process. It served as a sounding board to explore potential topics, brainstorm the initial concept of applying Agentic AI to the university admissions process, and sketch out a high-level project outline. Once the core idea was established, the independent research and development phases began.

2. Perplexity AI

- Product Name: Perplexity
- Source (URL): <https://www.perplexity.ai/>
- Software Functions Used: AI-assisted search engine and literature discovery.
- Extent of Use: Perplexity was utilized during the research phase to discover relevant academic papers, framework documentation, and related literature regarding Agentic AI and German university admissions. I independently read, evaluated, and properly cited all primary sources and scientific claims found through this tool.

3. Anthropic Claude (Sonnet)

- Product Name: Claude Sonnet (by Anthropic)
- Source (URL): <https://claude.ai/> (Accessed via the Antigravity IDE)
- Software Functions Used: Technical support for code implementation, debugging, algorithm optimization, and logic verification.
- Extent of Use: During the development of the Agentic AI prototype, Claude Sonnet was integrated within my IDE (Antigravity) to function as a technical programming assistant. It helped write code, debug errors, verify complex logical constraints, and optimize algorithmic efficiency. The underlying system architecture, agent orchestration, and core logic were designed independently by me. All AI-assisted code was rigorously reviewed, manually tested, and adapted to meet the specific requirements of the project.

4. Google Gemini

- Product Name: Google Gemini (Gemini Advanced)
- Source (URL): <https://gemini.google.com/>
- Software Functions Used: Natural language processing for linguistic refinement, grammar correction, stylistic smoothing, and detailed chapter structuring.
- Extent of Use: Gemini served primarily as a writing and editorial assistant during the drafting of the manuscript. I utilized the tool to refine my manually drafted paragraphs, improve sentence flow, correct grammatical errors, and elevate the text to a formal academic tone. It also aided in brainstorming logical structures for specific chapters (e.g., the Discussion and Future Work sections). The core scientific research, data analysis, and final conclusions were developed entirely independently.

All AI-generated suggestions were critically reviewed and verified for accuracy prior to integration into the final submission. I confirm that my intellectual and creative influence predominates in this work and that I maintained complete controlling authority over the use of these tools.

Declaration of authenticity

Surname:	Tran	First name:	Anh Tho
Matriculation no.:	332850	Degree program:	MSc. Digital Business Management

I hereby declare in lieu of oath that I have written this thesis independently and without the assistance of others. I have only used the sources and aids indicated and have identified the passages taken from them verbatim or in terms of content as such. I explicitly declare that Generative Artificial Intelligence tools were used for the purposes of initial ideation, linguistic refinement (grammar, stylistic smoothing) and technical support in coding implementation, including algorithm optimization and logic verification. I have verified all AI-generated suggestions, and the final content, argumentation, and scientific conclusions reflect my own intellectual work. The work has not been submitted in the same or a similar form to any other examination authority. I agree that the thesis may be checked for plagiarism with the help of a plagiarism detection service.



Pforzheim, 25/02/2026

Signature

Affidavit

"I, Anh Tho Tran, hereby declare that I have written this work entirely by myself. I marked any text, ideas, concepts, graphics, etc. taken directly or indirectly from external sources as such and provided complete references to the respective original sources. Furthermore, the work has not been submitted for examination at any other institution to date.

In addition, I declare... (one of the options must be selected and checked in consultation with the examiner):

☐ **Option 1: Permission to use AI-supported assistance tools without labeling requirement (supportive application)**

I hereby confirm that I have only used AI-supported assistance tools as support (e. g. spelling and grammar check, formatting, improvement of translations or my own wording) and that my intellectual and creative work is the predominant influence in this thesis. I am aware that the use of machine-generated content does not guarantee quality and accuracy, and I take full responsibility for any machine-generated passages I have used. I confirm that I have not used any AI tools which the examiner has explicitly excluded in writing or which are subject to labeling requirements.

☒ **Option 2: Mandatory labeling of AI-generated content if use is permitted**

I used artificial intelligence as an application and/or design tool, but I am aware that the use of machine-generated text, graphics, images, moving images, code, etc. does not guarantee quality and accuracy. I am fully responsible for the implementation of any machine-generated passages used by me and will be held liable in the event of incorrect content, breaches of data protection law, copyright or scientific misconduct (e.g. plagiarism). I hereby confirm that my intellectual and creative influence predominates in this work and that I have taken a controlling role in the use of AI-supported tools. I did not use any AI tools the examiner has explicitly excluded in writing. In addition, I confirm that I listed all AI-supported tools in the section "Overview of tools used" with their product name, my source (e. g., URL), and details on the software functions used and the extent of use. I have complied with any further documentation requirements specified by the examiners.

Note: If, at the time the assignment was issued, the responsible examiners have expressly stated that specific AI-supported writing tools are not subject to disclosure/labeling requirements, these do not need to be listed.

☐ **Option 3: Generative AI not permitted and not used**

I hereby confirm that I wrote this academic work completely on my own, i. e., I did not use any generative AI tools or assistance tools.”

A handwritten signature in black ink, consisting of stylized, cursive letters, positioned above a horizontal line.

Pforzheim, 25/02/2026

signature